Frozen GPT-2, ePC-trained and BP-trained reader caps: ordinary performance, dataset extremes and heavy tails (ext-20261009)

Capstan (pc_cap)

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Study `ext-20261009` · Capstan · 2026-10-20 · pc_cap code revision `a3727cf5e248` · commissioned by `docs/AgentHandoffForRese`
as amended by `docs/mandatory_ammendment.md`.

Provenance tags: [**record**] previously established in the frozen record and copied with its hashes; [**recomputed**] recomputed in this study from saved data; [**new**] newly evaluated in this study (own manifest `results/extremes_analysis/ext-20261009/manifest.json`). The frozen record was not modified.

1. Abstract and executive summary

Question. Across zsRE, CounterFact and MQuAKE, how do (i) frozen GPT-2 small without a cap, (ii) GPT-2 with the reader cap trained by error predictive coding (ePC) and (iii) GPT-2 with the same cap trained by backprop (BP) compare in ordinary benchmark performance, prediction difficulty, heavy-tail behaviour, extreme losses, adaptation benefit, correction magnitude, locality and stability — and does the ePC-trained cap show advantages in rare or extreme cases that survive a direct comparison with the BP-trained cap?

Design. The principal comparison is the frozen record's PC-reader study (family PCR): three paired training seeds per rule, identical architecture (3,348,228 reader parameters, taps and writes at blocks 4/8/12), identical initial parameters and training episodes within seed, adjoint acquisition in both arms, evaluated on realization 0 / order 100 with 300 edits on zsRE and CounterFact (checkpoints 100 and 300). This study adds [**new**]: a directly loaded frozen GPT-2 scored on exactly the same probes; teacher-forced target log-probabilities for all seven models on every probe (edit prompts, paraphrases, locality, near-miss, revision, unseen, MQuAKE multi-hop questions); the six saved readers evaluated on the sealed MQuAKE stream (6/6 complete at report time); write-vector magnitudes at every scored position; dataset frequency/entropy/shift statistics; and finite-range tail fits that start from a field-by-field reproduction of HT-17.

Headline findings.

- **zsre, paraphrase retention at 300 edits:** BP 97.3 %, 99.0 %, 97.7 %; ePC 94.7 %, 97.0 %, 98.0 % (seeds 0, 1, 2; 300 items each). Frozen: 0.0 %.
- **zsre, paraphrase per-token loss gain over frozen (seed mean):** BP 5.972 [5.708, 6.217]; ePC 5.895 [5.621, 6.146]; ePC − BP contrast -0.076 [-0.136, -0.027] nats/token (positive favours ePC).
- **counterfact, paraphrase retention at 300 edits:** BP 81.0 %, 80.0 %, 83.5 %; ePC 56.8 %, 77.8 %, 66.0 % (seeds 0, 1, 2; 300 items each). Frozen: 0.0 %.
- **counterfact, paraphrase per-token loss gain over frozen (seed mean):** BP 6.057 [5.807, 6.309]; ePC 4.941 [4.648, 5.228]; ePC − BP contrast -1.117 [-1.313, -0.916] nats/token (positive favours ePC).
- **mquake, paraphrase retention at 300 edits:** BP 62.3 %, 72.3 %, 67.7 %; ePC 60.0 %, 60.3 %, 58.3 % (seeds 0, 1, 2; 300 items each). Frozen: 0.0 %.
- **mquake, paraphrase per-token loss gain over frozen (seed mean):** BP 4.456 [4.193, 4.708]; ePC 3.961 [3.633, 4.261]; ePC − BP contrast -0.495 [-0.651, -0.347] nats/token (positive favours ePC).
- **Neither cap beats the frozen model everywhere.** Both caps convert the taught facts (own-prompt loss falls from ≈ 6 nats/token to ≈ 0.01) and both leave the frozen model's behaviour unchanged where they abstain; the differences between ePC- and BP-trained readers are differences of *gating*: which paraphrases and un-taught prompts trigger a write. Where a reader abstains, its loss equals the frozen loss to the bit.
- **ePC vs BP.** On CounterFact (frozen record) and on MQuAKE (this study's six supplemental evaluations) the ePC-trained reader retains fewer paraphrases than the BP-trained reader in all three seeds, and its per-probe paraphrase loss is higher in the paired contrast; on zsRE the two rules are close, with the seed-mean contrast slightly favouring BP and mixed per-seed signs. No analysis in this study (difficulty deciles, frozen-hard subsets, regressions, correction magnitudes) finds a regime of rare or extreme cases where the ePC-trained reader is reliably better than the BP-trained reader. Training cost remains 96–102× [**record**].
- **Extremes.** Frozen difficulty is large in the descriptive sense (per-token target surprisal has P99 above 10 nats on every dataset), but it is not heavy-tailed in the fitted sense: at a common absolute threshold (the frozen model's P95 of the pooled probe surprisal, same cases for every model) the generalized-Pareto shapes are: 19 fits, shapes -0.41 to 0.27; 4 intervals entirely below zero, 14 include zero, 1

entirely above zero (mquake ePC reader s2); at each model's own P95 they are: 21 fits, shapes -0.41 to 0.12; 5 intervals entirely below zero, 16 include zero, 0 entirely above zero — the own-quantile positives arise because a cap that drives many losses to zero lowers its own P95 into the frozen body, not because its extremes are heavier. The ordinary-text harm tails of the PC-reader cells reproduce HT-17 exactly (12/12 fields) and remain finite-range fits. Rare large per-probe deteriorations exist for both caps and occur almost only where a record fired (joint-extremes tail lift 14–62 on zsRE).

- **Qualifications.** One subject realization and one order; three training seeds are not three populations; GPT-2 small only; the frozen model answers essentially none of these prompts under the project's greedy convention, so the frozen baseline is informative through teacher-forced losses, not through exact-match rates.

2. Research questions and architectures

The cap is an episodic memory with a learned reader attached to a frozen GPT-2 small (12 blocks, 124,439,808 parameters, params digest `c4ac3fb8…`). Reads tap the residual stream after blocks 4, 8 and 12; writes add one vector per site at the current prediction position. Each edit is taught by five adjoint delta steps (lr 0.1, acceptance τ 0.1, aggregate budget A = 0.3) and appended as a record (key = prompt embedding at the three taps; payload = per-answer-token write deltas [3, 768]). At query time the reader scores the prompt against the nearest stored keys and keeps a null option; null mass ≥ 0.5 means no write, so the output equals the base to the bit. There is no inference-time iteration in either cap; the eight settling steps of ePC are a *training-time* quantity.

model	what differs	checkpoint	where evaluated
frozen GPT-2 small **[new]**	no reader, no memory, no writes	base params only	all probes, all datasets (Stage 2)
BP reader, seeds 0–2 **[record]**	reader trained by backprop (300 AdamW updates; feedforward answer CE, cached float16 teacher)	`train-bp-s*/theta_avg150-300.npz`	zsRE/CounterFact (record); MQuAKE (this study)
ePC reader, seeds 0–2 **[record]**	same episodes and initial parameters; base-dependent gradients from `EPCTrainer` (8 zero-initialised settling steps, rate 0.1, corrected SD-24 energy); 23.4–24.6 h vs 0.24–0.25 h	`train-epc-s*/theta_avg150-300.npz`	same

Verified differences between the two caps (handoff §1): architecture, parameter count, interface, acquisition rule, evaluation population and scoring code are identical; only the training-time gradient estimator for the base-dependent losses differs (settled ePC CE vs feedforward CE; float32 vs cached float16 teacher), hence the weights and the cost. The validation record `validation/gpu_checks.json` confirms: the directly loaded weights have the record's digest; a direct forward reproduces the saved cap-off losses to 0.000050 nats over 1524 positions; a reader with an empty memory, and a loaded memory on prompts where it abstains, returns the base logits exactly; the residual insertion arithmetic h_after = h_before + Δh holds at the site; saved RET-GS/RET-ES recompute exactly from the saved rows.

Families kept distinct and reported separately: Stage-4 `R1_learned_ff` (selected v5 BP-trained reader, 1,000 edits, 3 realizations $\times$ 5 orders) **[record]**; fixed-v5 acquisition credit (adjoint vs error inference, reader fixed) **[record]**; PC-v0 (regenerated ePC 50M base, v0 live cap) **[record, supplementary only]**.

3. Datasets and evaluation methodology

dataset	version used	pool	principal stream	probes per item	notes
zsRE	MEND release, train split rows; `zsre_eligible.jsonl` (10,420 eligible of 10,720 screened; 0 answered by the base)	10,420	realization 0, order 100, first 300 of 1,000	1 paraphrase, 1 NQ locality prompt	no relation ids; target = first `answers[]` entry
CounterFact	original CounterFact; `counterfact_eligible.jsonl` (20,091 of 20,391; 0 answered by the base)	20,091	same	2 paraphrases, ~10 neighbourhood prompts (first 3 scored)	`target_true` retained; relation ids
MQuAKE	**MQuAKE-CF** (9,218 cases; DEC-045); single-hop rewrite items + 80 composition cases per realization	6,043 items	300 items	1 question-form paraphrase, 2 locality prompts	`target_true`/`target_new` with QIDs; 3 multi-hop questions per case

Endpoint bundles per realization: 50 locality prompts, 100 near-miss pairs (distinct subjects within matched relation/template families; the reference is the cell's own cap-off neighbour response), 50 revision cases, 100 unseen prompts (un-taught pool items; false-fire measurement), and for MQuAKE 80 composition cases. Original metrics [**record**]: ES (immediate own-prompt success), RET-ES/RET-GS (end-of-stream own-prompt / paraphrase retention), LS (DEC-053 bounded exact decoded-text equality with the original-base response; *not* factual accuracy), near-miss preservation, revision, unseen false fires. Decoding: greedy, ≤ 32 new tokens, stop at newline/EOS; exact match of the NFKC/casefold/whitespace-normalised generation against supplied aliases.

Standardised measurements [**new**]: for prompt x_i and answer tokens y_i,1..T_i (the project's tokenisation: leading space, newline terminator), logP_i = $\sum$_t log p(y_i,t | x_i, y_i,<t) under teacher forcing with one prediction per gold prefix (the cap's writes act at every prediction position, exactly as the record's diagnostic); NLL_total_i = −logP_i; NLL_token_i = NLL_total_i / T_i (primary continuous difficulty measure); totals without the terminator are retained. Margins margin_total = logP(target_new) − logP(target_true) and margin_token = −NLL_token(new) + NLL_token(true). All seven models score the same serialised prompt/answer pairs; identical bucket padding; float64 log-softmax of float32 logits. Reader selection (record ids, null mass, hard null) and write norms ‖W_m‖ and ‖W_m‖/‖h_m‖ are recorded at every scored position. Probe counts: zsRE 1,350, CounterFact 2,300 (item-locality capped at 3 per item), MQuAKE 1,890 (incl. 240 composition questions).

Uncertainty: 2,000-draw percentile bootstrap resampling whole groups (items for item-level families, endpoint rows otherwise) with the same indices for every model (pairing preserved); windows for ordinary text; seeds reported individually and as a descriptive seed mean. These intervals cover within-population sampling only; no interval here speaks to new subjects or new training seeds.

4. Basic benchmark results

4.1 Registered metrics (original conventions, numerators/denominators)

zsre

edits	model	ES	RET-ES	RET-GS (primary)	LS	near-miss	unseen false fires	MQuAKE composition (all 3 questions)
300	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/300)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	
300	BP reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	97.3 % (292/300)	100.0 % (50/50)	85.0 % (85/100)	4/100	
300	BP reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	99.0 % (297/300)	100.0 % (50/50)	67.0 % (67/100)	14/100	
300	BP reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	97.7 % (293/300)	100.0 % (50/50)	73.0 % (73/100)	11/100	
300	ePC reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	94.7 % (284/300)	100.0 % (50/50)	91.0 % (91/100)	2/100	
300	ePC reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	97.0 % (291/300)	100.0 % (50/50)	85.0 % (85/100)	5/100	
300	ePC reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	98.0 % (294/300)	100.0 % (50/50)	85.0 % (85/100)	4/100	
100	frozen GPT-2 [new]	0.0 % (0/100)	0.0 % (0/100)	0.0 % (0/100)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	
100	BP reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	96.0 % (96/100)	100.0 % (50/50)	80.0 % (80/100)	5/100	
100	BP reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	99.0 % (99/100)	100.0 % (50/50)	56.0 % (56/100)	13/100	
100	BP reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	96.0 % (96/100)	100.0 % (50/50)	66.0 % (66/100)	8/100	
100	ePC reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	92.0 % (92/100)	100.0 % (50/50)	88.0 % (88/100)	2/100	
100	ePC reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	92.0 % (92/100)	100.0 % (50/50)	80.0 % (80/100)	5/100	
100	ePC reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	95.0 % (95/100)	100.0 % (50/50)	82.0 % (82/100)	4/100	

counterfact

edits	model	ES	RET-ES	RET-GS (primary)	LS	near-miss	unseen false fires	MQuAKE composition (all 3 questions)
300	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/300)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	
300	BP reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	81.0 % (243/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	
300	BP reader s1 [record]	99.0 % (297/300)	99.0 % (297/300)	80.0 % (240/300)	100.0 % (50/50)	99.0 % (99/100)	1/100	
300	BP reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	83.5 % (250.5/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	
300	ePC reader s0 [record]	100.0 % (300/300)	100.0 % (300/300)	56.8 % (170.5/300)	100.0 % (50/50)	98.0 % (98/100)	0/100	
300	ePC reader s1 [record]	100.0 % (300/300)	100.0 % (300/300)	77.8 % (233.5/300)	100.0 % (50/50)	95.0 % (95/100)	1/100	
300	ePC reader s2 [record]	100.0 % (300/300)	100.0 % (300/300)	66.0 % (198/300)	100.0 % (50/50)	99.0 % (99/100)	0/100	
100	frozen GPT-2 [new]	0.0 % (0/100)	0.0 % (0/100)	0.0 % (0/100)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	
100	BP reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	85.0 % (85/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	85.0 % (85/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	86.5 % (86.5/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	ePC reader s0 [record]	100.0 % (100/100)	100.0 % (100/100)	56.0 % (56/100)	100.0 % (50/50)	99.0 % (99/100)	0/100	
100	ePC reader s1 [record]	100.0 % (100/100)	100.0 % (100/100)	77.5 % (77.5/100)	100.0 % (50/50)	96.0 % (96/100)	0/100	
100	ePC reader s2 [record]	100.0 % (100/100)	100.0 % (100/100)	68.0 % (68/100)	100.0 % (50/50)	99.0 % (99/100)	0/100	

mquake

edits	model	ES	RET-ES	RET-GS (primary)	LS	near-miss	unseen false fires	MQuAKE composition (all 3 questions)
300	frozen GPT-2 [new]	0.0 % (0/300)	0.0 % (0/300)	0.0 % (0/300)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	question-level new-answer exact 0.0 %; old-answer exact 0.0 % (N/A as all-question success: no edits)
300	BP reader s0 [new]	100.0 % (300/300)	100.0 % (300/300)	62.3 % (187/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
300	BP reader s1 [new]	91.0 % (273/300)	94.0 % (282/300)	72.3 % (217/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
300	BP reader s2 [new]	100.0 % (300/300)	100.0 % (300/300)	67.7 % (203/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
300	ePC reader s0 [new]	100.0 % (300/300)	100.0 % (300/300)	60.0 % (180/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
300	ePC reader s1 [new]	100.0 % (300/300)	100.0 % (300/300)	60.3 % (181/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
300	ePC reader s2 [new]	100.0 % (300/300)	100.0 % (300/300)	58.3 % (175/300)	100.0 % (50/50)	100.0 % (100/100)	0/100	0.0 % (0/80)
100	frozen GPT-2 [new]	0.0 % (0/100)	0.0 % (0/100)	0.0 % (0/100)	100.0 % (50/50) (by definition)	100.0 % (100/100) (by definition)	N/A	question-level new-answer exact 0.0 %; old-answer exact 0.0 % (N/A as all-question success: no edits)
100	BP reader s0 [new]	100.0 % (100/100)	100.0 % (100/100)	62.0 % (62/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s1 [new]	86.0 % (86/100)	90.0 % (90/100)	68.0 % (68/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	BP reader s2 [new]	100.0 % (100/100)	100.0 % (100/100)	62.0 % (62/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	ePC reader s0 [new]	100.0 % (100/100)	100.0 % (100/100)	56.0 % (56/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	ePC reader s1 [new]	100.0 % (100/100)	100.0 % (100/100)	53.0 % (53/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	
100	ePC reader s2 [new]	100.0 % (100/100)	100.0 % (100/100)	51.0 % (51/100)	100.0 % (50/50)	100.0 % (100/100)	0/100	

Reading: the frozen model's ES/RET-ES/RET-GS are its greedy answers against the taught aliases — zero by the eligibility screen (every item was selected because the base did *not* answer it), and zero on paraphrases as well; its LS and near-miss are 1 by definition because its response *is* the reference. The caps reach own-prompt retention of 99–100 % and differ on paraphrases, near-miss preservation and false fires.

4.2 Standardised per-token losses (nats/token; mean over probes; 300-edit memories; frozen has no memory)

dataset	family	target	frozen	BP s0	BP s1	BP s2	ePC s0	ePC s1	ePC s2
zsre	edit	new	6.009 (n=300)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)
zsre	paraphrase	new	6.098 (n=300)	0.161 (n=300, fire 98 %)	0.077 (n=300, fire 99 %)	0.142 (n=300, fire 98 %)	0.320 (n=300, fire 95 %)	0.168 (n=300, fire 97 %)	0.120 (n=300, fire 98 %)
zsre	locality_item	true	6.223 (n=296)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)	6.223 (n=296, fire 0 %)
zsre	near_miss_neighbour	true	5.974 (n=100)	6.192 (n=100, fire 8 %)	6.161 (n=100, fire 21 %)	6.141 (n=100, fire 16 %)	6.048 (n=100, fire 3 %)	6.147 (n=100, fire 8 %)	6.106 (n=100, fire 7 %)
zsre	unseen	new	6.128 (n=100)	6.229 (n=100, fire 4 %)	6.209 (n=100, fire 14 %)	6.281 (n=100, fire 11 %)	6.209 (n=100, fire 2 %)	6.328 (n=100, fire 5 %)	6.289 (n=100, fire 4 %)
counterfact	edit	new	6.956 (n=300)	0.019 (n=300, fire 100 %)	0.100 (n=300, fire 99 %)	0.019 (n=300, fire 100 %)	0.019 (n=300, fire 100 %)	0.019 (n=300, fire 100 %)	0.019 (n=300, fire 100 %)
counterfact	paraphrase	new	7.369 (n=600)	1.362 (n=600, fire 86 %)	1.449 (n=600, fire 84 %)	1.125 (n=600, fire 88 %)	3.183 (n=600, fire 59 %)	1.598 (n=600, fire 81 %)	2.504 (n=600, fire 68 %)
counterfact	locality_item	true	6.295 (n=900)	6.288 (n=900, fire 0 %)	6.287 (n=900, fire 0 %)	6.292 (n=900, fire 0 %)	6.269 (n=900, fire 1 %)	6.272 (n=900, fire 2 %)	6.288 (n=900, fire 0 %)
counterfact	near_miss_neighbour	true	7.167 (n=100)	7.167 (n=100, fire 0 %)	7.148 (n=100, fire 1 %)	7.167 (n=100, fire 0 %)	7.110 (n=100, fire 2 %)	7.091 (n=100, fire 5 %)	7.157 (n=100, fire 1 %)
counterfact	unseen	new	6.924 (n=100)	6.924 (n=100, fire 0 %)	6.920 (n=100, fire 1 %)	6.924 (n=100, fire 0 %)	6.924 (n=100, fire 0 %)	6.920 (n=100, fire 1 %)	6.924 (n=100, fire 0 %)
counterfact	edit	true	6.382 (n=300)	5.215 (n=300, fire 100 %)	5.237 (n=300, fire 99 %)	5.215 (n=300, fire 100 %)	5.215 (n=300, fire 100 %)	5.215 (n=300, fire 100 %)	5.215 (n=300, fire 100 %)
mquake	edit	new	5.171 (n=300)	0.015 (n=300, fire 100 %)	0.342 (n=300, fire 94 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)	0.015 (n=300, fire 100 %)
mquake	paraphrase	new	6.048 (n=300)	2.009 (n=300, fire 72 %)	1.276 (n=300, fire 86 %)	1.491 (n=300, fire 83 %)	2.099 (n=300, fire 71 %)	1.994 (n=300, fire 72 %)	2.168 (n=300, fire 69 %)
mquake	locality_item	true	4.480 (n=560)	4.484 (n=560, fire 1 %)	4.495 (n=560, fire 1 %)	4.499 (n=560, fire 1 %)	4.490 (n=560, fire 1 %)	4.508 (n=560, fire 1 %)	4.501 (n=560, fire 1 %)
mquake	near_miss_neighbour	true	5.204 (n=100)	5.204 (n=100, fire 0 %)	5.204 (n=100, fire 0 %)	5.204 (n=100, fire 0 %)	5.204 (n=100, fire 0 %)	5.204 (n=100, fire 0 %)	5.204 (n=100, fire 0 %)
mquake	unseen	new	5.284 (n=100)	5.284 (n=100, fire 0 %)	5.284 (n=100, fire 0 %)	5.284 (n=100, fire 0 %)	5.284 (n=100, fire 0 %)	5.284 (n=100, fire 0 %)	5.284 (n=100, fire 0 %)
mquake	edit	true	4.579 (n=300)	8.010 (n=300, fire 100 %)	7.804 (n=300, fire 94 %)	8.010 (n=300, fire 100 %)	8.010 (n=300, fire 100 %)	8.010 (n=300, fire 100 %)	8.010 (n=300, fire 100 %)
mquake	composition	new	6.611 (n=240)	6.749 (n=240, fire 22 %)	6.813 (n=240, fire 43 %)	6.886 (n=240, fire 37 %)	6.746 (n=240, fire 19 %)	6.703 (n=240, fire 18 %)	6.716 (n=240, fire 20 %)
mquake	composition	old	5.489 (n=240)	5.770 (n=240, fire 22 %)	6.115 (n=240, fire 43 %)	6.037 (n=240, fire 37 %)	5.762 (n=240, fire 19 %)	5.700 (n=240, fire 18 %)	5.710 (n=240, fire 20 %)

4.3 Counterfactual preference (probes with both `target_new` and `target_true`)

dataset	family	model	n	mean margin_total (nats)	P(margin_total > 0)	mean margin_token	P(margin_token > 0)
counterfact	edit	BP reader s0	300	10.392	100.0 %	5.196	100.0 %
counterfact	edit	BP reader s1	300	10.274	99.0 %	5.137	99.0 %
counterfact	edit	BP reader s2	300	10.392	100.0 %	5.196	100.0 %
counterfact	edit	ePC reader s0	300	10.392	100.0 %	5.196	100.0 %
counterfact	edit	ePC reader s1	300	10.392	100.0 %	5.196	100.0 %
counterfact	edit	ePC reader s2	300	10.392	100.0 %	5.196	100.0 %
counterfact	edit	frozen GPT-2	300	-1.149	30.3 %	-0.574	30.3 %
counterfact	paraphrase	BP reader s0	600	7.264	86.7 %	3.632	86.7 %
counterfact	paraphrase	BP reader s1	600	7.122	85.2 %	3.561	85.2 %
counterfact	paraphrase	BP reader s2	600	7.532	88.3 %	3.766	88.3 %
counterfact	paraphrase	ePC reader s0	600	4.716	69.3 %	2.358	69.3 %
counterfact	paraphrase	ePC reader s1	600	6.930	84.3 %	3.465	84.3 %
counterfact	paraphrase	ePC reader s2	600	5.644	73.8 %	2.822	73.8 %
counterfact	paraphrase	frozen GPT-2	600	-1.146	31.2 %	-0.573	31.2 %
mquake	edit	BP reader s0	300	26.443	100.0 %	7.995	100.0 %
mquake	edit	BP reader s1	300	24.032	95.7 %	7.462	96.3 %
mquake	edit	BP reader s2	300	26.443	100.0 %	7.995	100.0 %
mquake	edit	ePC reader s0	300	26.443	100.0 %	7.995	100.0 %
mquake	edit	ePC reader s1	300	26.443	100.0 %	7.995	100.0 %
mquake	edit	ePC reader s2	300	26.443	100.0 %	7.995	100.0 %
mquake	edit	frozen GPT-2	300	-2.467	36.0 %	-0.592	45.0 %
mquake	paraphrase	BP reader s0	300	13.817	78.7 %	4.328	82.7 %
mquake	paraphrase	BP reader s1	300	17.112	90.0 %	5.224	91.0 %
mquake	paraphrase	BP reader s2	300	17.072	86.3 %	5.078	87.3 %
mquake	paraphrase	ePC reader s0	300	14.359	77.3 %	4.303	80.0 %
mquake	paraphrase	ePC reader s1	300	14.491	79.0 %	4.395	81.0 %
mquake	paraphrase	ePC reader s2	300	14.063	76.3 %	4.210	79.7 %
mquake	paraphrase	frozen GPT-2	300	-2.988	27.7 %	-0.758	37.3 %

Frozen GPT-2's original-fact performance: the frozen exact rate on `target_true` for CounterFact/MQuAKE

edit prompts is in `benchmark_original.csv` (`original_fact_exact_edit`); under the project's newline-stop greedy convention the base almost never emits a complete answer, so the teacher-forced `true`-target loss (table 4.2, family `edit`, target `true`) is the informative measure of what the base knew.

4.4 Context from the other families [record]

Stage-4 selected v5 reader (1,000 edits; 300 MQuAKE; three realizations × five orders): final RET-GS 96.03 % zsRE, 67.80 % CounterFact, 71.56 % MQuAKE; all 45 learned-reader cells exceed mean KL 0.001; registered classifier labels unchanged (`docs/R1_stage4_report.md`). Fixed-v5 acquisition credit (realization 0, order 100, 300 edits): paraphrase retention 98.3 %/98.3 % (zsRE adjoint/error) and 80.7 %/80.3 % (CounterFact); error credit raised the single worst ordinary-text token loss (12.27 vs 9.26 nats on zsRE) at ≈ 35 % more acquisition time (`docs/additional_work/PC-v1_report.md`). PC-v0 corrected credit on the **ePC 50M base** (not GPT-2 small): zsRE own-prompt retention +2.0 points, paraphrase −0.3 with mixed signs, CounterFact saturated (`docs/additional_work/PC-v0_report.md`). PC-reader training cost ratios ePC/BP: 102.1, 95.8, 96.7 (seeds 0, 1, 2).

5. Statistical characterisation of the datasets [new]

dataset	population	cases	K relations	H_rel	N_eff rel	K new targets	H_target	N_eff target	singleton targets	top-10 % target share	K subjects	singleton subjects	H(target | rel)	prompt tok med	prompt p99	answer tok med	answer p99	dup prompts
zsre	eligible	10420	n/a	n/a	n/a	5704	7.886	2660.5	44.7 %	46.2 %	10420	100.0 %	n/a	11	18	3	10	0
zsre	reader_train	1000	n/a	n/a	n/a	799	6.471	646.4	71.8 %	28.0 %	1000	100.0 %	n/a	11	18	3	9	0
zsre	sealed_r0	1000	n/a	n/a	n/a	779	6.412	609.0	69.3 %	29.1 %	1000	100.0 %	n/a	11	18	3	11	0
zsre	pcr_r0_300	300	n/a	n/a	n/a	262	5.470	237.5	81.3 %	21.7 %	300	100.0 %	n/a	11	19	3	11	0
counterfact	eligible	20091	34	3.409	30.2	738	5.556	258.8	0.5 %	58.2 %	20091	100.0 %	3.016	8	14	2	2	0
counterfact	reader_train	1000	34	3.379	29.3	288	5.140	170.7	12.4 %	42.9 %	1000	100.0 %	2.284	8	15	2	2	0
counterfact	sealed_r0	1000	34	3.383	29.5	291	5.210	183.1	12.0 %	39.1 %	1000	100.0 %	2.390	8	15	2	2	0
counterfact	pcr_r0_300	300	32	3.299	27.1	156	4.800	121.5	30.0 %	30.3 %	300	100.0 %	1.825	8	14	2	2	0
mquake	eligible	6043	37	3.074	21.6	2308	6.683	798.4	27.4 %	56.6 %	5577	85.8 %	3.811	9	16	3	11	0
mquake	reader_train	500	37	3.072	21.6	318	5.424	226.7	51.2 %	36.8 %	500	100.0 %	2.453	8	16	3	11	0
mquake	sealed_r0	300	34	2.928	18.7	213	5.113	166.2	59.3 %	32.0 %	300	100.0 %	2.229	9	15	3	9	0
mquake	pcr_r0_300	300	34	2.928	18.7	213	5.113	166.2	59.3 %	32.0 %	300	100.0 %	2.229	9	15	3	9	0

Populations: `eligible` (screened pool), `reader_train` (the first 1,000/1,000/500 items of the reader-training pools actually used), `sealed_r0` (the realization-0 stream), `pcr_r0_300` (its first 300 items = the PC-reader horizon). zsRE has no relation labels, so relation entropy is not manufactured. Zipf slopes over ranks 1–100 and all ranks are in `dataset_statistics.json`; a straight log-log segment is not a power-law test. Subject sets are disjoint between reader-training and sealed streams by construction (JS = ln 2 for subjects), so "rare in the benchmark" is a property of the draw, not of the world.

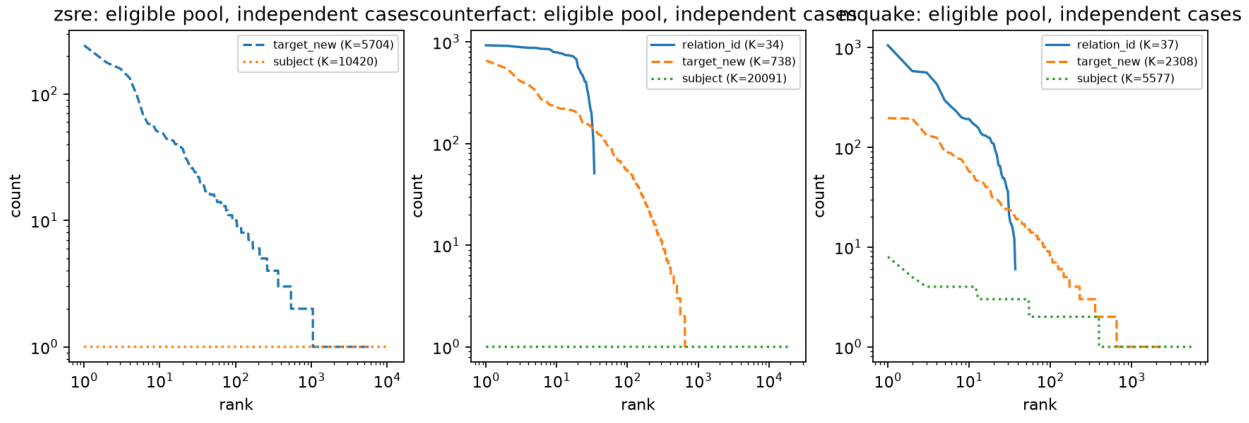


Figure 1: Rank-frequency of relation, new-target and subject categories in each eligible pool

5.1 Train/test shift (reader-training pool vs sealed realization 0)

dataset	variable (reader-train vs sealed r0)	JS (nats)	KL train‖test (α=0.5)	test categories unseen in train	median train count of test category	test categories with train count ≤ 1
zsre	target_new	0.4884	0.476	68.4 %	None	77.4 %
zsre	subject	0.6931	0.549	100.0 %	None	100.0 %
zsre	(subject, relation, target) triple overlap	0.0000				
counterfact	relation_id	0.0064	0.025	0.0 %	None	0.0 %
counterfact	target_new	0.1290	0.255	14.3 %	None	24.3 %
counterfact	target_true	0.1289	0.268	15.0 %	None	24.0 %
counterfact	subject	0.6931	0.549	100.0 %	None	100.0 %
counterfact	(subject, relation, target) triple overlap	0.0000				
mquake	relation_id	0.0240	0.083	0.0 %	None	1.0 %
mquake	target_new	0.4045	0.428	51.0 %	None	67.3 %
mquake	target_true	0.3032	0.345	39.0 %	None	45.3 %
mquake	subject	0.6931	0.481	100.0 %	None	100.0 %
mquake	(subject, relation, target) triple overlap	0.0000				

MQuAKE-CF raw: 9,218 cases; hop counts {'2': 4879, '3': 2535, '4': 1804}; requested edits per case {'1': 3755, '2': 3745, '3': 1282, '4': 436}; 577 distinct relation chains (H = 5.093, N_eff = 162.9); questions per case {'3': 9218}. Sealed r0 composition endpoint: 80 cases, hop counts {'2': 61, '3': 16, '4': 3}, edits per case {'1': 78, '2': 2}.

5.2 Baseline surprisal

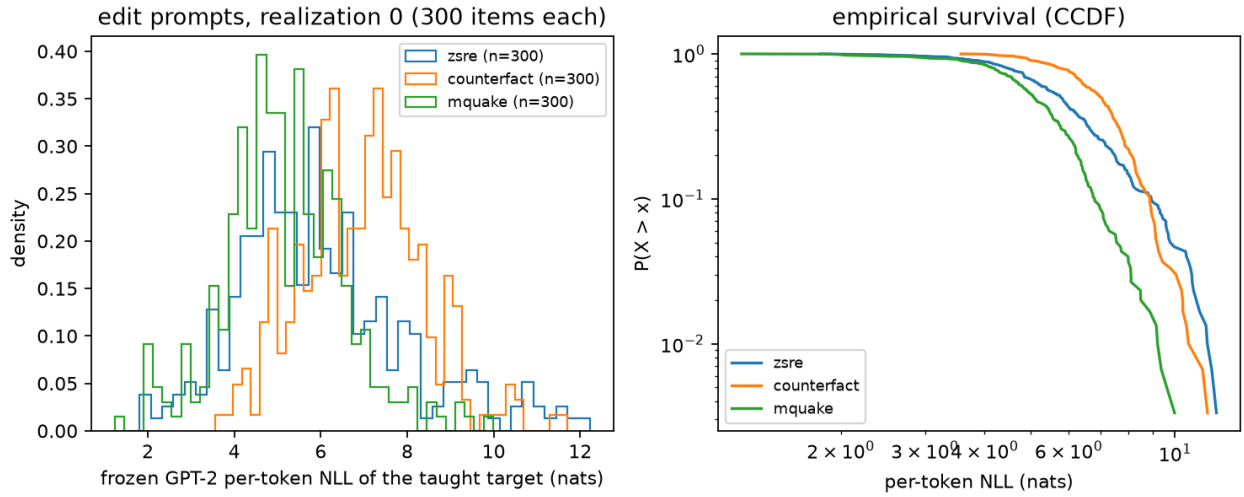


Figure 2: Frozen GPT-2 per-token target surprisal on edit prompts

Frozen GPT-2 per-token surprisal on the 245,237 ordinary-text positions (loss_capoff of a saved vector; identical in every cell): mean 4.080, median 3.468, P90 8.604, P95 10.129, P99 13.014, max 30.102, CVaR95 11.900 nats.

quantile	u	excesses	fit	κ	window-bootstrap 95 % κ	σ_u	GPD – exp (nats/excess)
0.900	8.60	24524	ok	-0.082	[-0.100, -0.071]	2.157	0.0053
0.950	10.13	12262	ok	-0.046	[-0.065, -0.028]	1.852	0.0016
0.975	11.44	6131	ok	-0.004	[-0.032, 0.022]	1.637	0.0000

6. Heavy tails and extreme-value statistics

Conventions (amendment §5): harm Δ = NLL(cap) − NLL(reference) at the same prefix on 1,931 windows / 245,237 ordinary-text positions, context reset per window; frequency P($\Delta > 0.01$), conditional severity, mean signed Δ, positive-part ES99+ including zero mass, maximum, KL(reference‖cap); fits of positive excesses with the record's analyser (`aw/tail_class.py`: ≥ 100 excesses in ≥ 30 windows, 200 joint-window bootstrap draws, five-fold held-out windows). Shapes are finite-range fits.

6.1 HT-17 reproduced [recomputed]

The twelve PC-reader cells of HT-17 were re-analysed from the saved vectors with the identical window plan: all twelve agree field by field with the record (`tables/ht17_reproduction_checks.csv`).

dataset	reader	mean Δ	P(Δ>0.01)	severity	Δ>0.01	ES99+	max Δ	GPD fit (u=0.01)	shape ξ	window-bootstrap 95 % ξ	GPD − exp (held-out nats/excess)
counterfact	bp s0	0.00202	0.0926 %	2.236	0.2070	8.52	eligible	-0.237	[-0.366, -0.110]	0.0177	
counterfact	bp s1	0.00535	0.2312 %	2.384	0.5513	12.50	eligible	-0.073	[-0.172, 0.025]	0.0002	
counterfact	bp s2	0.00118	0.0563 %	2.146	0.1208	9.86	eligible	-0.017	[-0.196, 0.136]	-0.0025	
counterfact	epc s0	0.00128	0.0661 %	2.005	0.1324	7.71	eligible	-0.262	[-0.460, -0.187]	-0.0120	
counterfact	epc s1	0.00628	0.2593 %	2.473	0.6413	12.36	eligible	-0.125	[-0.216, -0.043]	0.0045	
counterfact	epc s2	0.00195	0.1187 %	1.737	0.2061	12.62	eligible	0.059	[-0.083, 0.154]	-0.0028	
zsre	bp s0	0.00011	0.0061 %	1.913	0.0117	6.51	not_identified	—	—	—	
zsre	bp s1	0.00013	0.0049 %	2.603	0.0127	7.92	not_identified	—	—	—	
zsre	bp s2	0.00018	0.0077 %	2.366	0.0183	10.58	not_identified	—	—	—	
zsre	epc s0	0.00000	0.0000 %	—	0.0000	0.00	not_identified	—	—	—	
zsre	epc s1	0.00005	0.0016 %	2.971	0.0048	4.45	not_identified	—	—	—	
zsre	epc s2	0.00000	0.0000 %	—	0.0000	0.00	not_identified	—	—	—	

Stage-4 and PC-reader group rows of the record at u = 0.01 (joint-window intervals):

phase	dataset	condition	cells	P(Δ>0.01)	joint-window 95 %	severity	Δ>0.01	95 %
PC-reader	counterfact	bp	3	0.1267 %	[0.1108 %, 0.1435 %]	2.313	[2.134, 2.494]	
PC-reader	counterfact	epc	3	0.1480 %	[0.1311 %, 0.1653 %]	2.206	[1.996, 2.388]	
PC-reader	zsre	bp	3	0.0063 %	[0.0043 %, 0.0084 %]	2.280	[1.670, 2.969]	
PC-reader	zsre	epc	3	0.0005 %	[0.0001 %, 0.0011 %]	2.971	—	
stage4	counterfact	R1_learned_ff	15	0.2984 %	[0.2699 %, 0.3312 %]	1.913	[1.788, 2.038]	
stage4	counterfact	R1_nonlearned	15	2.0450 %	[1.9808 %, 2.1014 %]	3.486	[3.431, 3.543]	
stage4	counterfact	S1_LM	15	0.0000 %	[0.0000 %, 0.0000 %]	—	—	
stage4	counterfact	matched_update	15	0.0000 %	[0.0000 %, 0.0000 %]	—	—	
stage4	counterfact	v0_live_C1	15	0.0000 %	[0.0000 %, 0.0000 %]	—	—	
stage4	counterfact	v0_live_C2	15	0.0000 %	[0.0000 %, 0.0000 %]	—	—	
stage4	counterfact	v0_stable	15	0.0000 %	[0.0000 %, 0.0000 %]	—	—	
stage4	zsre	R1_learned_ff	15	0.1594 %	[0.1366 %, 0.1826 %]	1.630	[1.492, 1.777]	
stage4	zsre	R1_nonlearned	15	0.0266 %	[0.0215 %, 0.0317 %]	1.806	[1.569, 2.040]	
stage4	zsre	S1_LM	15	0.1049 %	[0.0968 %, 0.1121 %]	1.720	[1.520, 1.958]	
stage4	zsre	S1_literal	15	0.1041 %	[0.0962 %, 0.1112 %]	1.724	[1.525, 1.945]	
stage4	zsre	matched_update	15	0.1053 %	[0.0976 %, 0.1119 %]	1.677	[1.492, 1.863]	
stage4	zsre	v0_live_C1	15	0.0959 %	[0.0880 %, 0.1030 %]	2.188	[2.012, 2.397]	
stage4	zsre	v0_live_C2	15	0.0187 %	[0.0160 %, 0.0217 %]	17.203	[15.705, 18.744]	
stage4	zsre	v0_stable	15	0.1041 %	[0.0960 %, 0.1110 %]	1.716	[1.519, 1.933]	

6.2 MQuAKE PC-reader harm [new]

dataset	reader	mean Δ	P(Δ>0.01)	severity	Δ>0.01	ES99+	max Δ	GPD fit (u=0.01)	shape ξ	window-bootstrap 95 % ξ	GPD – exp (held-out nats/excess)
mquake	bp s0	0.00086	0.0351 %	2.518	0.0883	9.83	not_identified		—	—	
mquake	bp s1	0.00500	0.2589 %	1.981	0.5129	13.14	eligible	0.063	[-0.023, 0.190]	-0.0012	
mquake	bp s2	0.00086	0.0497 %	1.794	0.0893	7.77	eligible	-0.079	[-0.302, 0.112]	-0.0175	
mquake	epc s0	0.00011	0.0077 %	1.499	0.0116	4.53	not_identified		—	—	
mquake	epc s1	0.00555	0.2385 %	2.386	0.5691	12.08	eligible	-0.110	[-0.200, -0.005]	-0.0006	
mquake	epc s2	0.00115	0.0428 %	2.707	0.1159	11.19	eligible	-0.224	—	-0.0014	

Threshold sensitivity (all four registered thresholds):

reader	u	excesses	P(Δ>u)	severity	Δ>u	fit	ξ
bp s0	0.01	86	0.0351 %	2.518	not_identified	—	
bp s0	0.1	83	0.0338 %	2.606	not_identified	—	
bp s0	0.5	71	0.0290 %	3.001	not_identified	—	
bp s0	1.0	59	0.0241 %	3.436	not_identified	—	
bp s1	0.01	635	0.2589 %	1.981	eligible	0.063	
bp s1	0.1	600	0.2447 %	2.093	eligible	0.051	
bp s1	0.5	464	0.1892 %	2.621	eligible	-0.024	
bp s1	1.0	357	0.1456 %	3.179	eligible	-0.061	
bp s2	0.01	122	0.0497 %	1.794	eligible	-0.079	
bp s2	0.1	118	0.0481 %	1.853	eligible	-0.057	
bp s2	0.5	90	0.0367 %	2.339	not_identified	—	
bp s2	1.0	69	0.0281 %	2.823	not_identified	—	
epc s0	0.01	19	0.0077 %	1.499	not_identified	—	
epc s0	0.1	18	0.0073 %	1.579	not_identified	—	
epc s0	0.5	16	0.0065 %	1.749	not_identified	—	
epc s0	1.0	10	0.0041 %	2.401	not_identified	—	
epc s1	0.01	585	0.2385 %	2.386	eligible	-0.110	
epc s1	0.1	567	0.2312 %	2.460	eligible	-0.106	
epc s1	0.5	488	0.1990 %	2.811	eligible	-0.101	
epc s1	1.0	390	0.1590 %	3.331	eligible	-0.139	
epc s2	0.01	105	0.0428 %	2.707	eligible	-0.224	
epc s2	0.1	104	0.0424 %	2.732	eligible	-0.210	
epc s2	0.5	92	0.0375 %	3.047	not_identified	—	
epc s2	1.0	76	0.0310 %	3.536	not_identified	—	

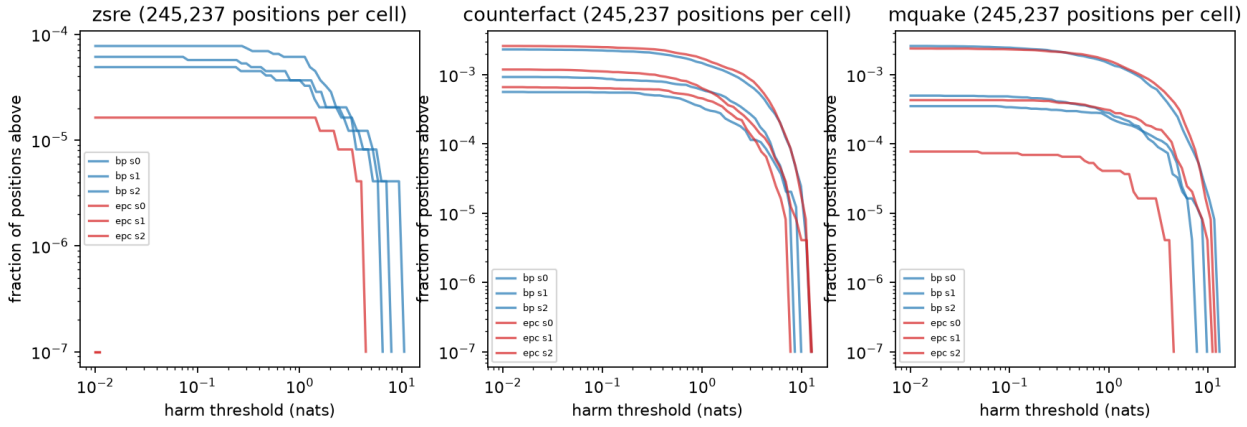


Figure 3: Harm survival, PC-reader cells

6.3 Per-probe target-loss tails [new]

Per-token NLL of the taught target on edit/paraphrase/unseen prompts; exceedances above the model's own P95; item-group bootstrap. Models that abstain share the frozen tail on those probes.

Probe-level (one value per probe; 300 edit items, 300–600 paraphrases, 100 unseen prompts): exceedances above P95 number 15–30, below the 50-excess floor, so no probe-level shape is reported — only the descriptive extremes.

dataset	family	model	n probes	mean	P95	P99	max	CVaR95	excesses > P95	fit
counterfact	edit	BP reader s0	300	0.02	0.05	0.06	0.06	0.05	15	insufficient
counterfact	edit	BP reader s1	300	0.10	0.05	0.13	9.43	1.57	15	insufficient
counterfact	edit	BP reader s2	300	0.02	0.05	0.06	0.06	0.05	15	insufficient
counterfact	edit	ePC reader s0	300	0.02	0.05	0.06	0.06	0.05	15	insufficient
counterfact	edit	ePC reader s1	300	0.02	0.05	0.06	0.06	0.05	15	insufficient
counterfact	edit	ePC reader s2	300	0.02	0.05	0.06	0.06	0.05	15	insufficient
counterfact	edit	frozen GPT-2	300	6.96	9.12	10.58	11.70	10.06	15	insufficient
counterfact	paraphrase	BP reader s0	600	1.36	8.19	9.32	10.50	9.00	30	insufficient
counterfact	paraphrase	BP reader s1	600	1.45	8.37	9.72	11.70	9.19	30	insufficient
counterfact	paraphrase	BP reader s2	600	1.13	7.78	9.13	10.83	8.69	30	insufficient
counterfact	paraphrase	ePC reader s0	600	3.18	9.11	9.89	11.56	9.66	30	insufficient
counterfact	paraphrase	ePC reader s1	600	1.60	8.40	9.53	15.56	9.32	30	insufficient
counterfact	paraphrase	ePC reader s2	600	2.50	8.86	10.38	11.93	9.69	30	insufficient
counterfact	paraphrase	frozen GPT-2	600	7.37	9.78	10.54	11.93	10.40	30	insufficient
counterfact	unseen	BP reader s0	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	BP reader s1	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	BP reader s2	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	ePC reader s0	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	ePC reader s1	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	ePC reader s2	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
counterfact	unseen	frozen GPT-2	100	6.92	9.45	9.73	10.01	9.70	5	insufficient
mquake	edit	BP reader s0	300	0.01	0.04	0.05	0.06	0.05	15	insufficient
mquake	edit	BP reader s1	300	0.34	3.90	7.25	8.47	5.78	15	insufficient
mquake	edit	BP reader s2	300	0.01	0.04	0.05	0.06	0.05	15	insufficient
mquake	edit	ePC reader s0	300	0.01	0.04	0.05	0.06	0.05	15	insufficient
mquake	edit	ePC reader s1	300	0.01	0.04	0.05	0.06	0.05	15	insufficient
mquake	edit	ePC reader s2	300	0.01	0.04	0.05	0.06	0.05	15	insufficient
mquake	edit	frozen GPT-2	300	5.17	7.53	9.15	9.99	8.39	15	insufficient
mquake	paraphrase	BP reader s0	300	2.01	7.12	8.59	11.06	8.09	15	insufficient
mquake	paraphrase	BP reader s1	300	1.28	5.92	8.59	10.45	7.22	15	insufficient
mquake	paraphrase	BP reader s2	300	1.49	6.50	7.70	10.45	7.38	15	insufficient
mquake	paraphrase	ePC reader s0	300	2.10	7.15	8.59	11.06	8.03	15	insufficient
mquake	paraphrase	ePC reader s1	300	1.99	6.85	8.06	10.45	7.77	15	insufficient
mquake	paraphrase	ePC reader s2	300	2.17	7.15	8.59	10.45	7.98	15	insufficient
mquake	paraphrase	frozen GPT-2	300	6.05	9.31	10.91	12.27	10.12	15	insufficient
mquake	unseen	BP reader s0	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	BP reader s1	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	BP reader s2	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	ePC reader s0	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	ePC reader s1	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	ePC reader s2	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
mquake	unseen	frozen GPT-2	100	5.28	7.21	7.91	10.18	7.98	5	insufficient
zsre	edit	BP reader s0	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	BP reader s1	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	BP reader s2	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	ePC reader s0	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	ePC reader s1	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	ePC reader s2	300	0.01	0.04	0.05	0.06	0.04	15	insufficient
zsre	edit	frozen GPT-2	300	6.01	9.71	11.63	12.22	10.91	15	insufficient
zsre	paraphrase	BP reader s0	300	0.16	0.07	6.03	6.43	2.70	15	insufficient
zsre	paraphrase	BP reader s1	300	0.08	0.05	0.26	6.43	1.13	15	insufficient
zsre	paraphrase	BP reader s2	300	0.14	0.06	5.73	6.43	2.34	15	insufficient
zsre	paraphrase	ePC reader	300	0.32	2.17	6.44	7.93	5.63	15	insufficient

Token-level (one value per target token, terminator excluded; groups = items for the bootstrap). `pooled_probes` pools edit, paraphrase and unseen prompts (new target) with item-locality, sealed locality and near-miss neighbour prompts (true target) and is the only population that clears the 100-excess screen. Two threshold rules are shown for it: each model's *own* P95 (the handoff's rule; a cap that has driven many losses to zero has a much lower P95, so its 'tail' starts in the frozen model's body) and the *common* absolute threshold u = frozen P95 (the same cases for every model, the comparison that answers the three-model question). Per-family rows are own-P90 and exploratory or insufficient:

dataset	family	model	threshold rule	n tokens	P99	max	threshold	excesses	fit	κ	item-bootstrap 95 % κ	σ_u	GPD − exp (in-sample nats/excess)
counterfact	locality_item	BP readers0	own quantile	900	14.69	17.54	P90 = 10.66	90	exploratory	-0.191	[-0.42, 0.06]	2.234	0.0113
counterfact	locality_item	BP readers1	own quantile	900	14.69	17.54	P90 = 10.66	90	exploratory	-0.191	[-0.42, 0.06]	2.234	0.0113
counterfact	locality_item	BP readers2	own quantile	900	14.69	17.54	P90 = 10.66	90	exploratory	-0.191	[-0.42, 0.06]	2.234	0.0113
counterfact	locality_item	EC readers0	own quantile	900	14.72	17.54	P90 = 10.70	90	exploratory	-0.198	[-0.45, 0.05]	2.299	0.0117
counterfact	locality_item	EC readers1	own quantile	900	14.90	19.00	P90 = 10.76	90	exploratory	-0.162	[-0.43, 0.05]	2.346	0.0091
counterfact	locality_item	EC readers2	own quantile	900	14.69	17.54	P90 = 10.66	90	exploratory	-0.191	[-0.42, 0.06]	2.234	0.0113
counterfact	locality_item	frozen GPT-2	own quantile	900	14.69	17.54	P90 = 10.66	90	exploratory	-0.191	[-0.42, 0.06]	2.234	0.0113
counterfact	paraphrase	BP readers0	own quantile	600	13.35	18.25	P90 = 8.67	60	exploratory	-0.241	[-0.50, -0.06]	3.271	0.0259
counterfact	paraphrase	BP readers1	own quantile	600	13.24	15.73	P90 = 8.86	60	exploratory	-0.460	[-1.10, -0.31]	3.548	0.0799
counterfact	paraphrase	BP readers2	own quantile	600	12.07	18.25	P90 = 7.57	60	exploratory	-0.192	[-0.91, -0.06]	2.895	0.0264
counterfact	paraphrase	EC readers0	own quantile	600	12.91	17.22	P90 = 10.73	60	exploratory	-0.027	[-1.13, 0.14]	1.222	0.0005
counterfact	paraphrase	EC readers1	own quantile	600	13.62	31.11	P90 = 8.80	60	exploratory	0.049	[-0.48, 0.22]	2.698	0.0025
counterfact	paraphrase	EC readers2	own quantile	600	14.33	17.22	P90 = 9.98	60	exploratory	-0.206	[-0.43, -0.01]	2.414	0.0172
counterfact	paraphrase	frozen GPT-2	own quantile	600	14.33	16.45	P90 = 11.99	60	exploratory	0.035	[-0.21, 0.35]	0.968	0.0004
counterfact	pooled_probes	BP readers0	frozen quantile (common)	2050	14.57	18.93	P95 = 12.70	69	exploratory	-0.008	[-0.37, 0.29]	1.514	0.0000
counterfact	pooled_probes	BP readers1	frozen quantile (common)	2050	14.52	18.93	P95 = 12.70	69	exploratory	-0.057	[-0.37, 0.29]	1.472	0.0010
counterfact	pooled_probes	BP readers2	frozen quantile (common)	2050	14.40	18.93	P95 = 12.70	60	exploratory	-0.034	[-0.29, 0.24]	1.538	0.0004
counterfact	pooled_probes	EC readers0	frozen quantile (common)	2050	14.49	18.93	P95 = 12.70	67	exploratory	0.005	[-0.31, 0.35]	1.434	0.0000
counterfact	pooled_probes	EC readers1	frozen quantile (common)	2050	14.68	31.11	P95 = 12.70	75	exploratory	0.193	[-0.30, 0.41]	1.435	0.0282
counterfact	pooled_probes	EC readers2	frozen quantile (common)	2050	14.65	18.93	P95 = 12.70	70	exploratory	-0.153	[-0.48, 0.13]	1.785	0.0072
counterfact	pooled_probes	frozen GPT-2	frozen quantile (common)	2050	14.85	18.93	P95 = 12.70	103	ok	-0.032	[-0.31, 0.23]	1.330	0.0003
counterfact	pooled_probes	BP readers0	own quantile	2050	14.57	18.93	P95 = 11.94	103	ok	-0.089	[-0.31, 0.10]	1.774	0.0029
counterfact	pooled_probes	BP readers1	own quantile	2050	14.52	18.93	P95 = 11.96	103	ok	-0.123	[-0.33, 0.06]	1.728	0.0062
counterfact	pooled_probes	BP readers2	own quantile	2050	14.40	18.93	P95 = 11.76	103	ok	-0.056	[-0.23, 0.14]	1.665	0.0012
counterfact	pooled_probes	EC readers0	own quantile	2050	14.49	18.93	P95 = 11.99	103	ok	-0.053	[-0.27, 0.12]	1.596	0.0010
counterfact	pooled_probes	EC readers1	own quantile	2050	14.68	31.11	P95 = 11.98	103	ok	0.088	[-0.31, 0.26]	1.743	0.0077
counterfact	pooled_probes	EC readers2	own quantile	2050	14.65	18.93	P95 = 11.97	103	ok	-0.161	[-0.36, 0.02]	1.922	0.0094
counterfact	pooled_probes	frozen GPT-2	own quantile	2050	14.85	18.93	P95 = 12.70	103	ok	-0.032	[-0.31, 0.23]	1.330	0.0003
mquake	locality_item	BP readers0	own quantile	1381	11.61	15.06	P90 = 8.65	138	ok	-0.235	[-0.60, -0.13]	1.817	0.0330
mquake	locality_item	BP readers1	own quantile	1381	11.64	22.50	P90 = 8.65	138	ok	-0.003	[-0.57, 0.11]	1.575	0.0000
mquake	locality_item	BP readers2	own quantile	1381	11.70	16.41	P90 = 8.68	138	ok	-0.164	[-0.56, -0.04]	1.802	0.0183
mquake	locality_item	EC readers0	own quantile	1381	11.64	22.50	P90 = 8.65	138	ok	0.002	[-0.57, 0.12]	1.550	0.0000
mquake	locality_item	EC readers1	own quantile	1381	11.89	22.50	P90 = 8.68	138	ok	-0.003	[-0.40, 0.11]	1.648	0.0000
mquake	locality_item	EC readers2	own quantile	1381	11.70	22.50	P90 = 8.65	138	ok	-0.007	[-0.54, 0.11]	1.597	0.0001
mquake	locality_item	frozen GPT-2	own quantile	1381	11.61	15.06	P90 = 8.65	138	ok	-0.229	[-0.59, -0.14]	1.791	0.0302
mquake	paraphrase	BP readers0	own quantile	755	17.87	23.13	P90 = 8.48	76	exploratory	-0.145	[-0.40, 0.04]	4.261	0.0071
mquake	paraphrase	BP reader	own quan-	755	13.41	17.65	P90 = 5.91	76	exploratory	-0.267	[-0.49, -0.07]	4.314	0.0224

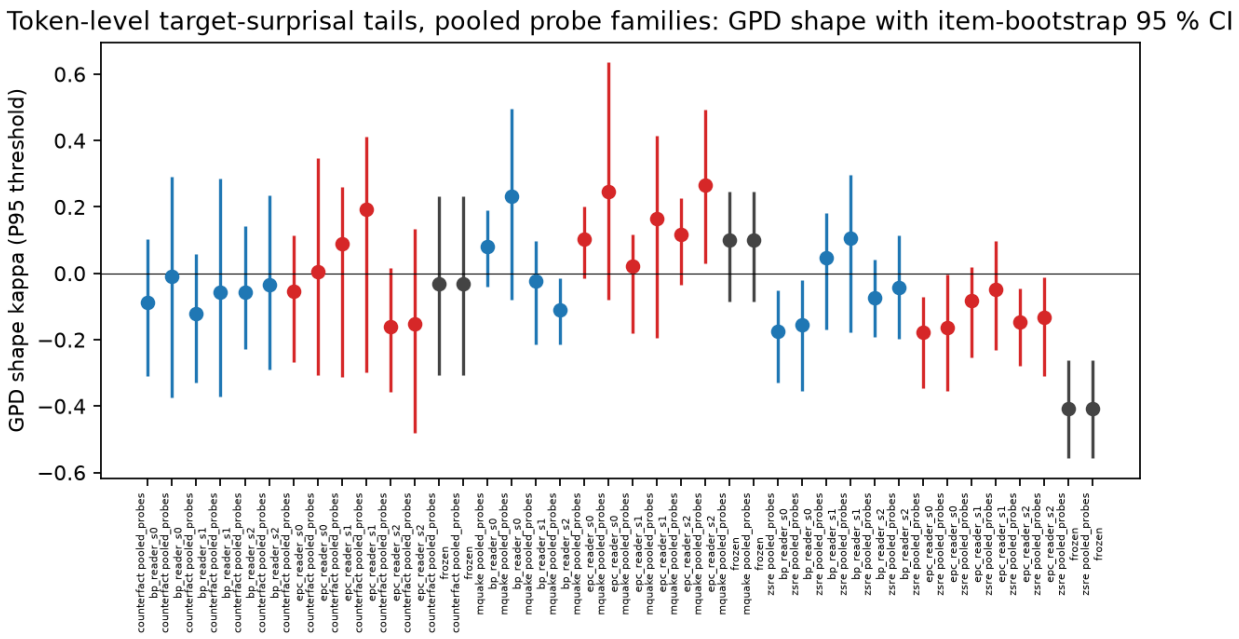


Figure 4: GPD shapes of per-probe target loss

7. Performance and adaptation in the extremes [new]

7.1 Paired gains over frozen GPT-2 and ePC vs BP

zsre (300 edits; 2,000 group-bootstrap draws, groups = items/rows)

family	target	gain	seed	n	mean	95 % CI	median	helped (G>0)	harmed (G<0)	mean of worst 5 % (−G)
edit	new	G_BP	0	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_BP	1	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_BP	2	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_BP	mean	300	5.995	[5.765, 6.211]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PC	0	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_PC	1	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_PC	2	300	5.995	[5.765, 6.211]	5.761	100.0 % [1.00, 1.00]	0.0 %	2.662
edit	new	G_PC	mean	300	5.995	[5.765, 6.211]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PCvsBP	0	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	1	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	2	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	mean	300	0.000	[0.000, 0.000]	—	0.0 % [0.00, 0.00]	—	—
paraphrase	new	G_BP	0	300	5.937	[5.666, 6.190]	5.695	97.3 % [0.95, 0.99]	0.3 %	1.114
paraphrase	new	G_BP	1	300	6.021	[5.762, 6.267]	5.720	99.0 % [0.98, 1.00]	0.3 %	2.009
paraphrase	new	G_BP	2	300	5.956	[5.689, 6.205]	5.699	97.7 % [0.96, 0.99]	0.3 %	1.297
paraphrase	new	G_BP	mean	300	5.972	[5.708, 6.217]	—	99.0 % [0.98, 1.00]	—	—
paraphrase	new	G_PC	0	300	5.778	[5.486, 6.048]	5.600	94.7 % [0.92, 0.97]	0.0 %	-0.000
paraphrase	new	G_PC	1	300	5.930	[5.654, 6.190]	5.699	97.0 % [0.95, 0.99]	0.3 %	0.933
paraphrase	new	G_PC	2	300	5.978	[5.711, 6.226]	5.701	98.0 % [0.96, 0.99]	0.0 %	1.577
paraphrase	new	G_PC	mean	300	5.895	[5.621, 6.146]	—	98.0 % [0.96, 0.99]	—	—
paraphrase	new	G_PCvsBP	0	300	-0.159	[-0.282, -0.055]	0.000	0.3 % [0.00, 0.01]	2.7 %	-3.082
paraphrase	new	G_PCvsBP	1	300	-0.092	[-0.176, -0.023]	0.000	0.0 % [0.00, 0.00]	2.0 %	-1.719
paraphrase	new	G_PCvsBP	2	300	0.021	[-0.048, 0.099]	0.000	1.0 % [0.00, 0.02]	0.7 %	-0.598
paraphrase	new	G_PCvsBP	mean	300	-0.076	[-0.136, -0.027]	—	0.3 % [0.00, 0.01]	—	—
unseen	new	G_BP	0	100	-0.102	[-0.243, 0.000]	0.000	0.0 % [0.00, 0.00]	4.0 %	-1.696
unseen	new	G_BP	1	100	-0.082	[-0.376, 0.243]	0.000	4.0 % [0.01, 0.08]	10.0 %	-4.044
unseen	new	G_BP	2	100	-0.154	[-0.435, 0.125]	0.000	2.0 % [0.00, 0.05]	9.0 %	-4.044
unseen	new	G_BP	mean	100	-0.112	[-0.328, 0.096]	—	4.0 % [0.01, 0.08]	—	—
unseen	new	G_PC	0	100	-0.082	[-0.220, 0.000]	0.000	0.0 % [0.00, 0.00]	2.0 %	-1.361
unseen	new	G_PC	1	100	-0.200	[-0.406, -0.039]	0.000	0.0 % [0.00, 0.00]	5.0 %	-3.339
unseen	new	G_PC	2	100	-0.162	[-0.354, -0.017]	0.000	0.0 % [0.00, 0.00]	4.0 %	-2.696
unseen	new	G_PC	mean	100	-0.148	[-0.310, -0.024]	—	0.0 % [0.00, 0.00]	—	—
unseen	new	G_PCvsBP	0	100	0.020	[0.000, 0.057]	0.000	2.0 % [0.00, 0.05]	0.0 %	-0.000
unseen	new	G_PCvsBP	1	100	-0.118	[-0.381, 0.093]	0.000	5.0 % [0.01, 0.10]	4.0 %	-3.450
unseen	new	G_PCvsBP	2	100	-0.008	[-0.235, 0.184]	0.000	5.0 % [0.01, 0.10]	2.0 %	-1.940
unseen	new	G_PCvsBP	mean	100	-0.035	[-0.189, 0.094]	—	7.0 % [0.02, 0.13]	—	—
locality_item	true	G_BP	0	296	0.000	[-0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
locality_item	true	G_BP	1	296	0.000	[-0.000, 0.000]	0.000	0.0 % [0.00,	0.0 %	-0.000

counterfact (300 edits; 2,000 group-bootstrap draws, groups = items/rows)

family	target	gain	seed	n	mean	95 % CI	median	helped (G>0)	harmed (G<0)	mean of worst 5 % (−G)
edit	new	G_BP	0	300	6.938	[6.774, 7.091]	6.986	100.0 % [1.00, 1.00]	0.0 %	4.314
edit	new	G_BP	1	300	6.857	[6.683, 7.027]	6.942	99.0 % [0.98, 1.00]	0.0 %	3.435
edit	new	G_BP	2	300	6.938	[6.774, 7.091]	6.986	100.0 % [1.00, 1.00]	0.0 %	4.314
edit	new	G_BP	mean	300	6.911	[6.750, 7.069]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PC	0	300	6.938	[6.774, 7.091]	6.986	100.0 % [1.00, 1.00]	0.0 %	4.314
edit	new	G_PC	1	300	6.938	[6.774, 7.091]	6.986	100.0 % [1.00, 1.00]	0.0 %	4.314
edit	new	G_PC	2	300	6.938	[6.774, 7.091]	6.986	100.0 % [1.00, 1.00]	0.0 %	4.314
edit	new	G_PC	mean	300	6.938	[6.774, 7.091]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PCvsBP	0	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	1	300	0.081	[0.000, 0.182]	0.000	1.0 % [0.00, 0.02]	0.0 %	-0.000
edit	new	G_PCvsBP	2	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	mean	300	0.027	[0.000, 0.061]	—	1.0 % [0.00, 0.02]	—	—
paraphrase	new	G_BP	0	600	6.007	[5.706, 6.282]	6.741	84.7 % [0.81, 0.88]	0.8 %	-0.058
paraphrase	new	G_BP	1	600	5.921	[5.628, 6.197]	6.679	83.8 % [0.80, 0.87]	0.2 %	-0.017
paraphrase	new	G_BP	2	600	6.244	[5.980, 6.515]	6.814	87.7 % [0.85, 0.91]	0.0 %	-0.000
paraphrase	new	G_BP	mean	600	6.057	[5.807, 6.309]	—	94.0 % [0.92, 0.96]	—	—
paraphrase	new	G_PC	0	600	4.186	[3.838, 4.555]	5.607	58.7 % [0.54, 0.63]	0.2 %	-0.040
paraphrase	new	G_PC	1	600	5.771	[5.464, 6.070]	6.687	80.8 % [0.77, 0.84]	0.5 %	-0.200
paraphrase	new	G_PC	2	600	4.865	[4.511, 5.205]	6.204	68.2 % [0.64, 0.72]	0.2 %	-0.040
paraphrase	new	G_PC	mean	600	4.941	[4.648, 5.228]	—	85.2 % [0.82, 0.89]	—	—
paraphrase	new	G_PCvsBP	0	600	-1.821	[-2.168, -1.447]	0.000	3.7 % [0.02, 0.06]	29.5 %	-9.353
paraphrase	new	G_PCvsBP	1	600	-0.149	[-0.357, 0.057]	0.000	5.2 % [0.04, 0.07]	8.7 %	-7.370
paraphrase	new	G_PCvsBP	2	600	-1.379	[-1.666, -1.085]	0.000	3.5 % [0.02, 0.05]	22.7 %	-9.211
paraphrase	new	G_PCvsBP	mean	600	-1.117	[-1.313, -0.916]	—	5.8 % [0.04, 0.08]	—	—
unseen	new	G_BP	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_BP	1	100	0.004	[0.000, 0.011]	0.000	1.0 % [0.00, 0.03]	0.0 %	-0.000
unseen	new	G_BP	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_BP	mean	100	0.001	[0.000, 0.004]	—	1.0 % [0.00, 0.03]	—	—
unseen	new	G_PC	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PC	1	100	0.004	[0.000, 0.011]	0.000	1.0 % [0.00, 0.03]	0.0 %	-0.000
unseen	new	G_PC	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PC	mean	100	0.001	[0.000, 0.004]	—	1.0 % [0.00, 0.03]	—	—
unseen	new	G_PCvsBP	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	1	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	mean	100	0.000	[0.000, 0.000]	—	0.0 % [0.00, 0.00]	—	—
locality_item	true	G_BP	0	900	0.007	[0.000, 0.016]	0.000	0.4 % [0.00, 0.01]	0.0 %	-0.000
locality_item	true	G_BP	1	900	0.009	[0.000, 0.020]	0.000	0.4 % [0.00, 0.01]	0.0 %	-0.000
locality_item	true	G_BP	2	900	0.003	[0.000, 0.010]	0.000	0.1 % [0.00, 0.00]	0.0 %	-0.000
locality_item	true	G_BP	mean	900	0.006	[0.001, 0.014]	—	0.7 % [0.00, 0.01]	—	—
locality_item	true	G_PC	0	900	0.027	[0.006, 0.050]	0.000	0.9 % [0.00,	0.1 %	-0.049

mquake (300 edits; 2,000 group-bootstrap draws, groups = items/rows)

family	target	gain	seed	n	mean	95 % CI	median	helped (G>0)	harmed (G<0)	mean of worst 5 % (−G)
edit	new	G_BP	0	300	5.157	[5.004, 5.311]	5.049	100.0 % [1.00, 1.00]	0.0 %	2.225
edit	new	G_BP	1	300	4.829	[4.621, 5.023]	4.956	94.0 % [0.91, 0.97]	0.0 %	-0.000
edit	new	G_BP	2	300	5.157	[5.004, 5.311]	5.049	100.0 % [1.00, 1.00]	0.0 %	2.225
edit	new	G_BP	mean	300	5.047	[4.890, 5.199]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PC	0	300	5.157	[5.004, 5.311]	5.049	100.0 % [1.00, 1.00]	0.0 %	2.225
edit	new	G_PC	1	300	5.157	[5.004, 5.311]	5.049	100.0 % [1.00, 1.00]	0.0 %	2.225
edit	new	G_PC	2	300	5.157	[5.004, 5.311]	5.049	100.0 % [1.00, 1.00]	0.0 %	2.225
edit	new	G_PC	mean	300	5.157	[5.004, 5.311]	—	100.0 % [1.00, 1.00]	—	—
edit	new	G_PCvsBP	0	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	1	300	0.328	[0.179, 0.485]	0.000	6.0 % [0.03, 0.09]	0.0 %	-0.000
edit	new	G_PCvsBP	2	300	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
edit	new	G_PCvsBP	mean	300	0.109	[0.060, 0.162]	—	6.0 % [0.03, 0.09]	—	—
paraphrase	new	G_BP	0	300	4.039	[3.708, 4.349]	4.785	71.7 % [0.66, 0.77]	0.0 %	-0.000
paraphrase	new	G_BP	1	300	4.772	[4.488, 5.021]	5.438	85.7 % [0.82, 0.89]	0.3 %	-0.058
paraphrase	new	G_BP	2	300	4.557	[4.269, 4.817]	5.276	82.3 % [0.78, 0.86]	0.3 %	-0.058
paraphrase	new	G_BP	mean	300	4.456	[4.193, 4.708]	—	89.3 % [0.86, 0.93]	—	—
paraphrase	new	G_PC	0	300	3.949	[3.621, 4.263]	4.711	70.7 % [0.65, 0.75]	0.0 %	-0.000
paraphrase	new	G_PC	1	300	4.054	[3.713, 4.359]	4.819	72.3 % [0.67, 0.77]	0.0 %	-0.000
paraphrase	new	G_PC	2	300	3.880	[3.543, 4.194]	4.690	69.0 % [0.64, 0.74]	0.3 %	-0.098
paraphrase	new	G_PC	mean	300	3.961	[3.633, 4.261]	—	75.7 % [0.71, 0.80]	—	—
paraphrase	new	G_PCvsBP	0	300	-0.090	[-0.299, 0.106]	0.000	4.3 % [0.02, 0.07]	5.3 %	-5.871
paraphrase	new	G_PCvsBP	1	300	-0.718	[-0.943, -0.502]	0.000	0.7 % [0.00, 0.02]	13.7 %	-6.652
paraphrase	new	G_PCvsBP	2	300	-0.677	[-0.882, -0.486]	0.000	0.3 % [0.00, 0.01]	13.3 %	-6.515
paraphrase	new	G_PCvsBP	mean	300	-0.495	[-0.651, -0.347]	—	4.0 % [0.02, 0.06]	—	—
unseen	new	G_BP	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_BP	1	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_BP	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_BP	mean	100	-0.000	[-0.000, 0.000]	—	0.0 % [0.00, 0.00]	—	—
unseen	new	G_PC	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PC	1	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PC	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PC	mean	100	-0.000	[-0.000, 0.000]	—	0.0 % [0.00, 0.00]	—	—
unseen	new	G_PCvsBP	0	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	1	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	2	100	0.000	[0.000, 0.000]	0.000	0.0 % [0.00, 0.00]	0.0 %	-0.000
unseen	new	G_PCvsBP	mean	100	0.000	[0.000, 0.000]	—	0.0 % [0.00, 0.00]	—	—
locality_item	true	G_BP	0	560	-0.004	[-0.011, 0.002]	0.000	0.2 % [0.00, 0.01]	0.4 %	-0.084
locality_item	true	G_BP	1	560	-0.015	[-0.040, 0.000]	0.000	0.0 % [0.00, 0.00]	0.5 %	-0.282
locality_item	true	G_BP	2	560	-0.019	[-0.040, -0.002]	0.000	0.2 % [0.00, 0.01]	0.9 %	-0.449
locality_item	true	G_BP	mean	560	-0.012	[-0.024, -0.003]	—	0.4 % [0.00, 0.01]	—	—
locality_item	true	G_PC	0	560	-0.010	[-0.037, 0.009]	0.000	0.2 % [0.00,	0.5 %	-0.282

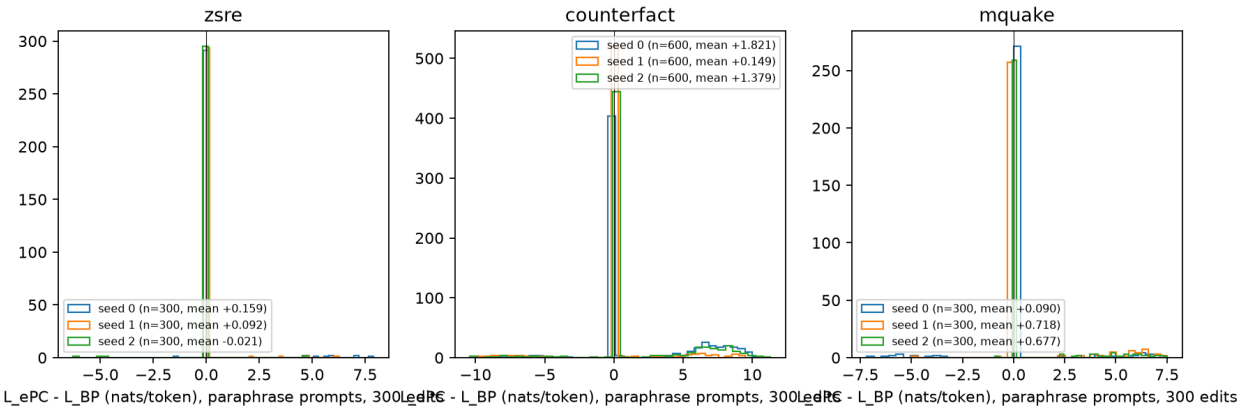


Figure 5: Paired ePC − BP differences on paraphrases

7.2 Gain versus frozen difficulty (deciles fixed by the frozen per-token NLL)

dataset	family	decile	n	frozen NLL range	G_BP (seed mean)	G_PC (seed mean)	G_PCvsBP
counterfact	edit	1	30	3.57–4.97	4.581 [4.45, 4.71]	4.581 [4.45, 4.71]	0.000 [0.00, 0.00]
counterfact	edit	5	30	6.51–6.98	6.751 [6.70, 6.81]	6.751 [6.70, 6.81]	0.000 [0.00, 0.00]
counterfact	edit	10	30	8.86–11.70	9.434 [9.06, 9.76]	9.538 [9.27, 9.85]	0.104 [0.00, 0.42]
counterfact	paraphrase	1	60	3.62–5.51	3.941 [3.45, 4.36]	3.222 [2.68, 3.73]	-0.719 [-1.05, -0.38]
counterfact	paraphrase	5	60	6.86–7.38	5.727 [5.08, 6.25]	4.492 [3.75, 5.15]	-1.234 [-1.79, -0.67]
counterfact	paraphrase	10	60	9.39–11.93	8.580 [7.84, 9.16]	7.221 [6.41, 8.09]	-1.359 [-2.05, -0.69]
counterfact	unseen	1	10	3.30–4.82	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]
counterfact	unseen	5	10	6.59–6.87	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]
counterfact	unseen	10	10	8.99–10.01	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]
mquake	edit	1	30	1.24–3.55	2.649 [2.42, 2.89]	2.716 [2.48, 2.94]	0.067 [0.00, 0.17]
mquake	edit	5	30	4.80–5.06	4.800 [4.63, 4.92]	4.910 [4.88, 4.94]	0.110 [0.00, 0.27]
mquake	edit	10	30	6.80–9.99	7.404 [7.02, 7.82]	7.757 [7.46, 8.07]	0.354 [0.08, 0.64]
mquake	paraphrase	1	30	1.44–3.83	2.445 [2.01, 2.79]	2.196 [1.65, 2.62]	-0.249 [-0.53, 0.01]
mquake	paraphrase	5	30	5.59–6.06	3.823 [2.90, 4.70]	3.819 [2.87, 4.70]	-0.004 [-0.33, 0.26]
mquake	paraphrase	10	30	8.57–12.27	6.654 [5.85, 7.31]	6.637 [5.79, 7.30]	-0.017 [-0.24, 0.19]
mquake	unseen	1	10	1.72–3.12	-0.000 [-0.00, 0.00]	-0.000 [-0.00, 0.00]	0.000 [0.00, 0.00]
mquake	unseen	5	10	5.12–5.37	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]
mquake	unseen	10	10	6.85–10.18	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]	0.000 [0.00, 0.00]
zsre	edit	1	30	1.81–3.75	3.080 [2.87, 3.27]	3.080 [2.87, 3.27]	0.000 [0.00, 0.00]
zsre	edit	5	30	5.26–5.76	5.510 [5.46, 5.56]	5.510 [5.46, 5.56]	0.000 [0.00, 0.00]
zsre	edit	10	30	8.92–12.22	10.163 [9.83, 10.48]	10.163 [9.83, 10.48]	0.000 [0.00, 0.00]
zsre	paraphrase	1	30	1.92–3.93	2.959 [2.53, 3.26]	2.965 [2.64, 3.22]	0.006 [-0.14, 0.18]
zsre	paraphrase	5	30	5.37–5.74	5.359 [4.97, 5.58]	5.296 [4.87, 5.56]	-0.062 [-0.22, 0.00]
zsre	paraphrase	10	30	9.25–12.69	10.637 [10.35, 10.94]	10.637 [10.35, 10.94]	0.000 [0.00, 0.00]
zsre	unseen	1	10	2.41–3.67	-0.192 [-0.89, 0.51]	-0.110 [-0.33, 0.00]	0.082 [-0.62, 0.80]
zsre	unseen	5	10	5.39–6.07	0.202 [0.00, 0.61]	0.000 [0.00, 0.00]	-0.202 [-0.61, 0.00]
zsre	unseen	10	10	8.62–12.08	-1.214 [-2.35, -0.25]	-0.945 [-2.08, 0.00]	0.269 [0.00, 0.68]

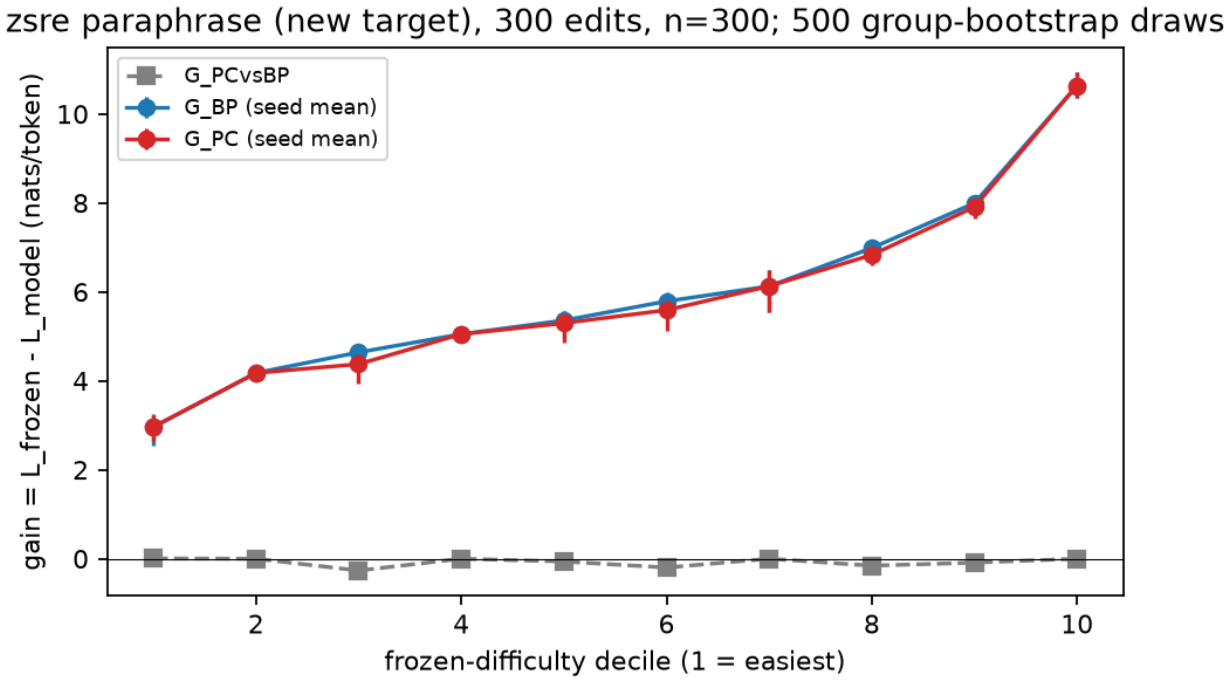


Figure 6: Deciles, zsRE paraphrase

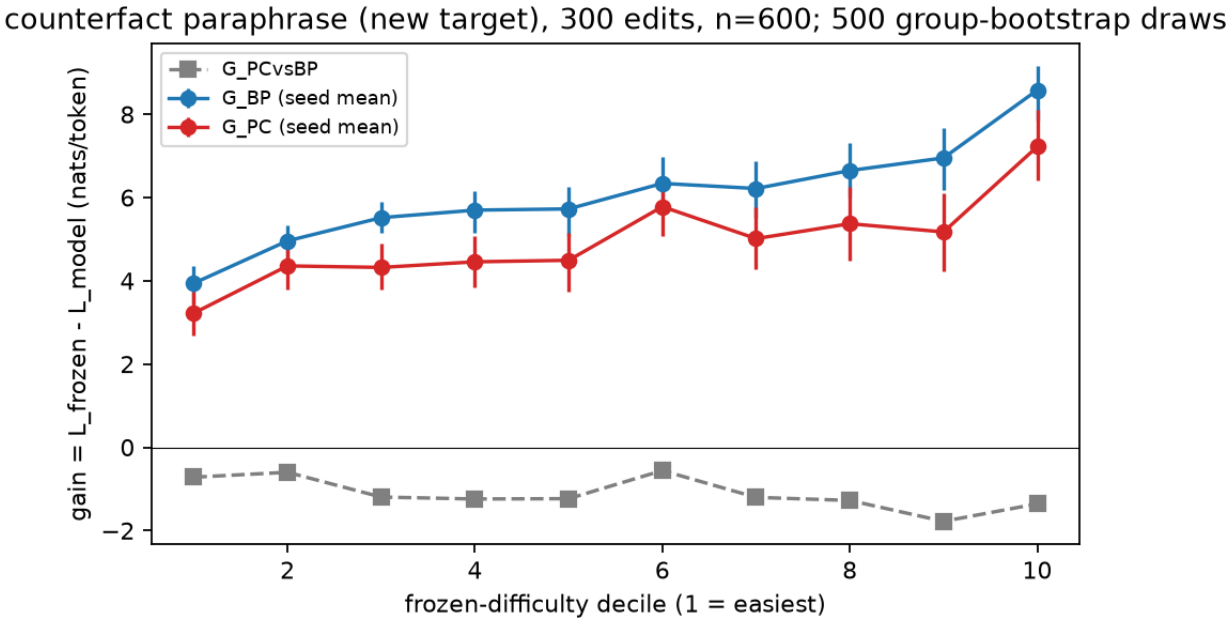


Figure 7: Deciles, CounterFact paraphrase

7.3 Two worst-case summaries (A: each model's own worst 5 %; B: the frozen model's hardest 5 %, fixed)

dataset	family	model	n	mean loss	A: own CVaR95	k	B: mean loss on frozen-hard 5 %	B: success on frozen-hard 5 %	B: mean loss on frozen-hard 1 %
counterfact	edit	BP s0	300	0.019	0.054	15	0.018	100.0 %	0.008
counterfact	edit	BP s1	300	0.100	1.568	15	0.644	93.3 %	0.008
counterfact	edit	BP s2	300	0.019	0.054	15	0.018	100.0 %	0.008
counterfact	edit	ePC s0	300	0.019	0.054	15	0.018	100.0 %	0.008
counterfact	edit	ePC s1	300	0.019	0.054	15	0.018	100.0 %	0.008
counterfact	edit	ePC s2	300	0.019	0.054	15	0.018	100.0 %	0.008
counterfact	edit	frozen GPT-2	300	6.956	10.063	15	10.126	0.0 %	11.271
counterfact	paraphrase	BP s0	600	1.362	8.998	30	2.217	76.7 %	1.944
counterfact	paraphrase	BP s1	600	1.449	9.191	30	2.003	80.0 %	4.171
counterfact	paraphrase	BP s2	600	1.125	8.689	30	1.933	80.0 %	3.744
counterfact	paraphrase	ePC s0	600	3.183	9.660	30	2.908	73.3 %	2.097
counterfact	paraphrase	ePC s1	600	1.598	9.317	30	2.028	80.0 %	4.815
counterfact	paraphrase	ePC s2	600	2.504	9.690	30	3.408	70.0 %	7.722
counterfact	paraphrase	frozen GPT-2	600	7.369	10.396	30	10.416	0.0 %	11.394
counterfact	unseen	BP s0	100	6.924	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	BP s1	100	6.920	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	BP s2	100	6.924	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	ePC s0	100	6.924	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	ePC s1	100	6.920	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	ePC s2	100	6.924	9.699	5	9.752	0.0 %	10.008
counterfact	unseen	frozen GPT-2	100	6.924	9.699	5	9.752	0.0 %	10.008
mquake	edit	BP s0	300	0.015	0.047	15	0.020	100.0 %	0.025
mquake	edit	BP s1	300	0.342	5.779	15	1.660	80.0 %	0.025
mquake	edit	BP s2	300	0.015	0.047	15	0.020	100.0 %	0.025
mquake	edit	ePC s0	300	0.015	0.047	15	0.020	100.0 %	0.025
mquake	edit	ePC s1	300	0.015	0.047	15	0.020	100.0 %	0.025
mquake	edit	ePC s2	300	0.015	0.047	15	0.020	100.0 %	0.025
mquake	edit	frozen GPT-2	300	5.171	8.392	15	8.450	0.0 %	9.517
mquake	paraphrase	BP s0	300	2.009	8.086	15	3.484	33.3 %	6.048
mquake	paraphrase	BP s1	300	1.276	7.218	15	2.960	33.3 %	3.427
mquake	paraphrase	BP s2	300	1.491	7.378	15	2.960	33.3 %	3.427
mquake	paraphrase	ePC s0	300	2.099	8.032	15	3.484	33.3 %	6.048
mquake	paraphrase	ePC s1	300	1.994	7.771	15	2.960	33.3 %	3.427
mquake	paraphrase	ePC s2	300	2.168	7.981	15	2.960	33.3 %	3.427
mquake	paraphrase	frozen GPT-2	300	6.048	10.116	15	10.170	0.0 %	11.528
mquake	unseen	BP s0	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	BP s1	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	BP s2	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	ePC s0	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	ePC s1	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	ePC s2	100	5.284	7.985	5	8.141	0.0 %	10.184
mquake	unseen	frozen GPT-2	100	5.284	7.985	5	8.141	0.0 %	10.184
zsre	edit	BP s0	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	BP s1	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	BP s2	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	ePC s0	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	ePC s1	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	ePC s2	300	0.015	0.043	15	0.022	100.0 %	0.008
zsre	edit	frozen GPT-2	300	6.009	10.908	15	10.988	0.0 %	11.952
zsre	paraphrase	BP s0	300	0.161	2.695	15	0.046	100.0 %	0.045
zsre	paraphrase	BP s1	300	0.077	1.130	15	0.046	100.0 %	0.045
zsre	paraphrase	BP s2	300	0.142	2.338	15	0.046	100.0 %	0.045
zsre	paraphrase	ePC s0	300	0.320	5.635	15	0.046	100.0 %	0.045
zsre	paraphrase	ePC s1	300	0.168	2.831	15	0.046	100.0 %	0.045
zsre	paraphrase	ePC s2	300	0.120	1.934	15	0.046	100.0 %	0.045
zsre	paraphrase	frozen GPT-2	300	6.098	11.331	15	11.365	0.0 %	12.332
zsre	unseen	BP s0	100	6.229	11.584	5	10.551	0.0 %	12.078
zsre	unseen	BP s1	100	6.209	12.712	5	11.323	0.0 %	12.078
zsre	unseen	BP s2	100	6.281	12.712	5	11.323	0.0 %	12.078
zsre	unseen	ePC s0	100	6.209	11.584	5	10.551	0.0 %	12.078
zsre	unseen	ePC s1	100	6.328	12.620	5	11.323	0.0 %	12.078
zsre	unseen	ePC s2	100	6.289	12.054	5	10.551	0.0 %	12.078
zsre	unseen	frozen GPT-2	100	6.128	10.353	5	10.551	0.0 %	12.078

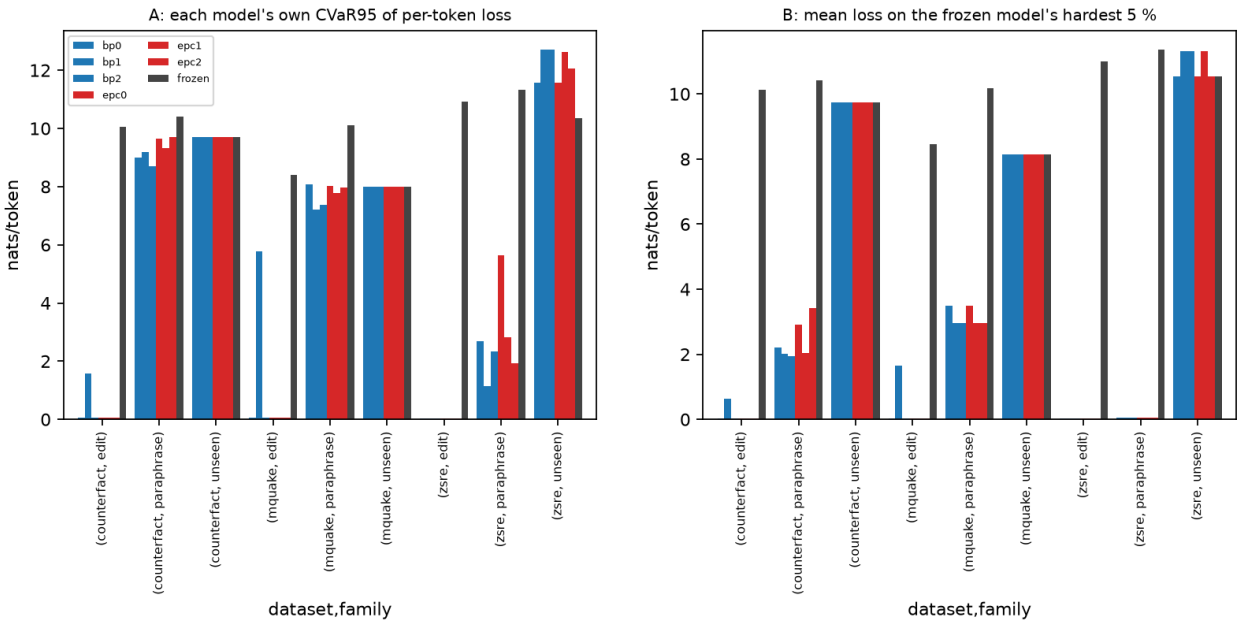


Figure 8: CVaR A and B

7.4 Rare and severe regressions (D = L_model − L_frozen; rows `epc{s}_minus_bp{s}` use L_BP as reference)

dataset	family	target	model (D = L_model − L_ref)	n	P(D>0)	mean D	D>0	P99 D	worst-5 % mean D	max D	P(D>0.5)	P(D>1)
counterfact	edit	new	BP s0	300	0.0 %	—	-4.079	-4.314	-3.564	0.00 %	0.00 %	
counterfact	edit	new	BP s1	300	0.0 %	—	-3.528	-3.435	0.000	0.00 %	0.00 %	
counterfact	edit	new	BP s2	300	0.0 %	—	-4.079	-4.314	-3.564	0.00 %	0.00 %	
counterfact	edit	new	ePC s0	300	0.0 %	—	-4.079	-4.314	-3.564	0.00 %	0.00 %	
counterfact	edit	new	epc0_minus_bp0	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	edit	new	ePC s1	300	0.0 %	—	-4.079	-4.314	-3.564	0.00 %	0.00 %	
counterfact	edit	new	epc1_minus_bp1	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	edit	new	ePC s2	300	0.0 %	—	-4.079	-4.314	-3.564	0.00 %	0.00 %	
counterfact	edit	new	epc2_minus_bp2	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	BP s0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	BP s1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	BP s2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	ePC s0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	epc0_minus_bp0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	ePC s1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	epc1_minus_bp1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	ePC s2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality	true	epc2_minus_bp2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality_item	true	BP s0	900	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality_item	true	BP s1	900	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality_item	true	BP s2	900	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality_item	true	ePC s0	900	0.1 %	2.264	0.000	0.049	2.264	0.11 %	0.11 %	
counterfact	locality_item	true	epc0_minus_bp0	900	0.4 %	1.407	0.000	0.122	2.264	0.44 %	0.33 %	
counterfact	locality_item	true	ePC s1	900	0.3 %	1.240	0.000	0.081	2.264	0.22 %	0.22 %	
counterfact	locality_item	true	epc1_minus_bp1	900	0.4 %	0.987	0.000	0.086	2.264	0.22 %	0.22 %	
counterfact	locality_item	true	ePC s2	900	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	locality_item	true	epc2_minus_bp2	900	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	BP s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	BP s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	BP s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	ePC s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	epc0_minus_bp0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	ePC s1	100	1.0 %	0.090	0.001	0.015	0.090	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	epc1_minus_bp1	100	2.0 %	1.029	0.109	0.343	1.969	1.00 %	1.00 %	
counterfact	near_miss_neighbour	true	ePC s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	near_miss_neighbour	true	epc2_minus_bp2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	paraphrase	new	BP s0	600	0.8 %	0.359	0.000	0.058	1.237	0.17 %	0.17 %	
counterfact	paraphrase	new	BP s1	600	0.2 %	0.523	0.000	0.017	0.523	0.17 %	0.00 %	
counterfact	paraphrase	new	BP s2	600	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	paraphrase	new	ePC s0	600	0.2 %	1.237	0.000	0.040	1.237	0.17 %	0.17 %	
counterfact	paraphrase	new	epc0_minus_bp0	600	29.5 %	6.987	9.587	9.353	10.516	28.67 %	28.50 %	
counterfact	paraphrase	new	ePC s1	600	0.5 %	2.069	0.000	0.200	3.865	0.50 %	0.50 %	
counterfact	paraphrase	new	epc1_minus_bp1	600	8.7 %	5.856	8.646	7.370	9.494	8.17 %	8.17 %	
counterfact	paraphrase	new	ePC s2	600	0.2 %	1.237	0.000	0.040	1.237	0.17 %	0.17 %	
counterfact	paraphrase	new	epc2_minus_bp2	600	22.7 %	7.032	9.704	9.211	11.317	22.33 %	22.33 %	
counterfact	unseen	new	BP s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	BP s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	BP s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	ePC s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	epc0_minus_bp0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	ePC s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	epc1_minus_bp1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	ePC s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
counterfact	unseen	new	epc2_minus_bp2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	edit	new	BP s0	300	0.0 %	—	-1.970	-2.225	-1.220	0.00 %	0.00 %	
mquake	edit	new	BP s1	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	edit	new	BP s2	300	0.0 %	—	-1.970	-2.225	-1.220	0.00 %	0.00 %	
mquake	edit	new	ePC s0	300	0.0 %	—	-1.970	-2.225	-1.220	0.00 %	0.00 %	
mquake	edit	new	epc0_minus_bp0	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	edit	new	ePC s1	300	0.0 %	—	-1.970	-2.225	-1.220	0.00 %	0.00 %	
mquake	edit	new	epc1_minus_bp1	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	edit	new	ePC s2	300	0.0 %	—	-1.970	-2.225	-1.220	0.00 %	0.00 %	
mquake	edit	new	epc2_minus_bp2	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	BP s0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	BP s1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	BP s2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	ePC s0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	epc0_minus_bp0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	ePC s1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	epc1_minus_bp1	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	ePC s2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality	true	epc2_minus_bp2	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	locality_item	true	BP s0	560	0.4 %	1.213	0.000	0.084	1.386	0.36 %	0.36 %	
mquake	locality_item	true	BP s1	560	0.5 %	2.727	0.000	0.282	5.755	0.54 %	0.54 %	
mquake	locality_item	true	BP s2	560	0.9 %	2.604	0.000	0.449	3.684	0.89 %	0.89 %	
mquake	locality_item	true	ePC s0	560	0.5 %	2.727	0.000	0.282	5.755	0.54 %	0.54 %	
mquake	locality_item	true	epc0_minus_bp0	560	0.4 %	3.096	0.000	0.214	5.755	0.18 %	0.18 %	
mquake	locality_item	true	ePC s1	560	1.1 %	3.129	0.426	0.647	5.755	1.07 %	1.07 %	
mquake	locality_item	true	epc1_minus_bp1	560	0.5 %	3.532	0.000	0.365	3.684	0.54 %	0.54 %	
mquake	locality_item	true	ePC s2	560	0.7 %	3.587	0.000	0.495	5.755	0.71 %	0.71 %	
mquake	locality_item	true	epc2_minus_bp2	560	0.2 %	5.755	0.000	0.198	5.755	0.18 %	0.18 %	
mquake	near_miss_neighbour	true	BP s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	BP s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	BP s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	ePC s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	epc0_minus_bp0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	ePC s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	epc1_minus_bp1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	ePC s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	near_miss_neighbour	true	epc2_minus_bp2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	paraphrase	new	BP s0	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	paraphrase	new	BP s1	300	0.3 %	0.926	0.000	0.058	0.926	0.33 %	0.00 %	
mquake	paraphrase	new	BP s2	300	0.3 %	0.926	0.000	0.058	0.926	0.33 %	0.00 %	
mquake	paraphrase	new	ePC s0	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	paraphrase	new	epc0_minus_bp0	300	5.3 %	5.871	6.639	5.871	7.184	5.33 %	5.33 %	
mquake	paraphrase	new	ePC s1	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	paraphrase	new	epc1_minus_bp1	300	13.7 %	5.399	7.129	6.652	7.517	13.67 %	13.67 %	
mquake	paraphrase	new	ePC s2	300	0.3 %	1.573	0.000	0.098	1.573	0.33 %	0.33 %	
mquake	paraphrase	new	epc2_minus_bp2	300	13.3 %	5.101	6.948	6.515	7.517	13.33 %	13.33 %	
mquake	unseen	new	BP s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	BP s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	BP s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	ePC s0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	epc0_minus_bp0	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	ePC s1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	epc1_minus_bp1	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	ePC s2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
mquake	unseen	new	epc2_minus_bp2	100	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
zsre	edit	new	BP s0	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	BP s1	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	BP s2	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	ePC s0	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	epc0_minus_bp0	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
zsre	edit	new	ePC s1	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	epc1_minus_bp1	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
zsre	edit	new	ePC s2	300	0.0 %	—	-2.208	-2.662	-1.792	0.00 %	0.00 %	
zsre	edit	new	epc2_minus_bp2	300	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	
zsre	locality	true	BP s0	50	0.0 %	—	0.000	0.000	0.000	0.00 %	0.00 %	

Worst fifteen regressions at 300 edits:

dataset	family	target	model	item	subject	D	L_frozen	L_model	fired	max write	rel.
zsre	near_miss_neighbour	true	BP s2	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	near_miss_neighbour	true	ePC s1	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	near_miss_neighbour	true	BP s0	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	near_miss_neighbour	true	BP s1	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	near_miss_neighbour	true	ePC s0	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	near_miss_neighbour	true	ePC s2	zsre:0:near:83	nan	7.75	6.84	14.59	True	0.121	
zsre	unseen	new	ePC s1	zsre-train-17050	Stefan Glarner	6.35	6.53	12.88	True	0.151	
zsre	unseen	new	ePC s2	zsre-train-17050	Stefan Glarner	6.35	6.53	12.88	True	0.151	
zsre	unseen	new	BP s2	zsre-train-17050	Stefan Glarner	6.35	6.53	12.88	True	0.151	
zsre	unseen	new	BP s1	zsre-train-17050	Stefan Glarner	6.35	6.53	12.88	True	0.151	
mquake	locality_item	true	ePC s0	mquake:a5a3061bcff3487fb9570	nan	7.75	6.20	11.95	True	0.120	
mquake	locality_item	true	BP s1	mquake:a5a3061bcff3487fb9570	nan	7.75	6.20	11.95	True	0.120	
mquake	locality_item	true	ePC s2	mquake:a5a3061bcff3487fb9570	nan	7.75	6.20	11.95	True	0.120	
mquake	locality_item	true	ePC s1	mquake:a5a3061bcff3487fb9570	nan	7.75	6.20	11.95	True	0.120	
zsre	unseen	new	BP s2	zsre-train-2345	Angela Petrelli	5.67	9.37	15.03	True	0.123	

8. CAP correction mechanisms [new]

Write magnitudes are the vectors actually added at the three sites at each scored prediction position (captured from the base's partial-pass call during `predict`), with ‖h_m‖ the pre-write residual row at the same position; the frozen model has structural zeros and is not fitted.

dataset	family	model	n	fired	mean max ‖W‖ site 1 (block 4)	site 2 (block 8)	site 3 (block 12)	mean max rel. site 1	site 2	site 3	P95 rel. (max over sites)	max rel.
zsre	edit	BP reader s0	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	BP reader s0	300	293	7.069	10.658	47.336	0.1070	0.1054	0.1234	0.1749	0.2097
zsre	edit	BP reader s1	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	BP reader s1	300	298	7.069	10.658	47.307	0.1071	0.1055	0.1234	0.1748	0.2097
zsre	edit	BP reader s2	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	BP reader s2	300	294	7.069	10.658	47.313	0.1071	0.1055	0.1233	0.1741	0.2097
zsre	edit	ePC reader s0	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	ePC reader s0	300	284	7.069	10.658	47.284	0.1070	0.1054	0.1230	0.1746	0.2097
zsre	edit	ePC reader s1	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	ePC reader s1	300	292	7.069	10.658	47.305	0.1070	0.1054	0.1233	0.1749	0.2097
zsre	edit	ePC reader s2	300	300	7.069	10.658	47.257	0.1075	0.1057	0.1232	0.1735	0.2306
zsre	paraphrase	ePC reader s2	300	294	7.069	10.658	47.280	0.1070	0.1054	0.1231	0.1748	0.2097
counterfact	edit	BP reader s0	300	300	7.066	10.658	44.890	0.1076	0.1043	0.1199	0.1409	0.1906
counterfact	paraphrase	BP reader s0	600	513	7.064	10.658	44.912	0.1110	0.1093	0.1176	0.1381	0.1620
counterfact	edit	BP reader s1	300	297	7.066	10.658	44.859	0.1076	0.1044	0.1199	0.1409	0.1906
counterfact	paraphrase	BP reader s1	600	504	7.065	10.658	44.842	0.1111	0.1094	0.1172	0.1377	0.1620
counterfact	edit	BP reader s2	300	300	7.066	10.658	44.890	0.1076	0.1043	0.1199	0.1409	0.1906
counterfact	paraphrase	BP reader s2	600	526	7.066	10.659	44.812	0.1111	0.1095	0.1173	0.1379	0.1620
counterfact	edit	ePC reader s0	300	300	7.066	10.658	44.890	0.1076	0.1043	0.1199	0.1409	0.1906
counterfact	paraphrase	ePC reader s0	600	353	7.063	10.658	45.017	0.1104	0.1086	0.1164	0.1381	0.1620
counterfact	edit	ePC reader s1	300	300	7.066	10.658	44.890	0.1076	0.1043	0.1199	0.1409	0.1906
counterfact	paraphrase	ePC reader s1	600	488	7.065	10.658	44.893	0.1110	0.1093	0.1174	0.1379	0.1620
counterfact	edit	ePC reader s2	300	300	7.066	10.658	44.890	0.1076	0.1043	0.1199	0.1409	0.1906
counterfact	paraphrase	ePC reader s2	600	410	7.064	10.658	45.096	0.1107	0.1088	0.1173	0.1389	0.1620
mquake	edit	BP reader s0	300	300	7.070	10.658	46.249	0.1116	0.1088	0.1362	0.1835	0.3564
mquake	paraphrase	BP reader s0	300	215	7.070	10.658	46.237	0.1071	0.1045	0.1271	0.1866	0.3582
mquake	edit	BP reader s1	300	282	7.070	10.658	45.739	0.1121	0.1088	0.1344	0.1783	0.3564
mquake	paraphrase	BP reader s1	300	258	7.070	10.658	46.293	0.1073	0.1045	0.1268	0.1839	0.3582
mquake	edit	BP reader s2	300	300	7.070	10.658	46.249	0.1116	0.1088	0.1362	0.1835	0.3564
mquake	paraphrase	BP reader s2	300	248	7.070	10.658	46.586	0.1073	0.1046	0.1288	0.1868	0.3582
mquake	edit	ePC reader s0	300	300	7.070	10.658	46.249	0.1116	0.1088	0.1362	0.1835	0.3564
mquake	paraphrase	ePC reader s0	300	212	7.070	10.658	46.444	0.1075	0.1048	0.1287	0.1874	0.3582
mquake	edit	ePC reader s1	300	300	7.070	10.658	46.249	0.1116	0.1088	0.1362	0.1835	0.3564
mquake	paraphrase	ePC reader s1	300	217	7.070	10.658	46.393	0.1074	0.1047	0.1283	0.1865	0.3582
mquake	edit	ePC reader s2	300	300	7.070	10.658	46.249	0.1116	0.1088	0.1362	0.1835	0.3564
mquake	paraphrase	ePC reader s2	300	208	7.070	10.658	46.453	0.1075	0.1049	0.1291	0.1868	0.3582

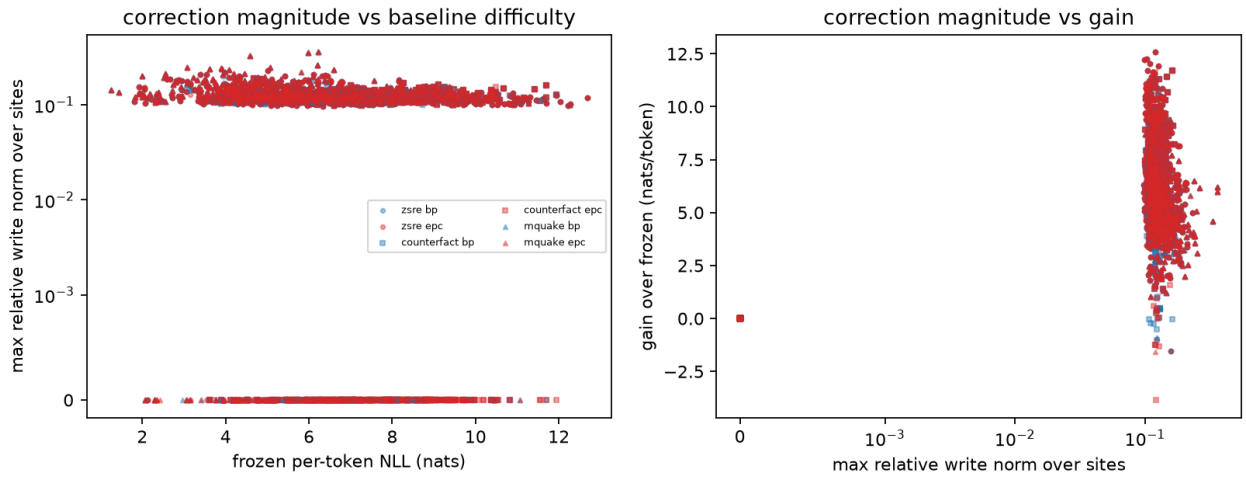


Figure 9: Write norms vs difficulty and gain

Rank correlations (Spearman) and joint extremes:

dataset	family	model	n	fired	ρ(frozen NLL, write)	ρ(write, gain)	ρ(frozen NLL, gain)	ρ(write, gain) fired only
counterfact	edit	BP s0	300	300	+0.15 (p=0.0081)	+0.15 (p=0.0078)	+1.00 (p=0)	+0.15 (p=0.0078)
counterfact	edit	BP s1	300	297	+0.14 (p=0.015)	+0.18 (p=0.0014)	+0.96 (p=9.6e-175)	+0.16 (p=0.006)
counterfact	edit	BP s2	300	300	+0.15 (p=0.0081)	+0.15 (p=0.0078)	+1.00 (p=0)	+0.15 (p=0.0078)
counterfact	edit	ePC s0	300	300	+0.15 (p=0.0081)	+0.15 (p=0.0078)	+1.00 (p=0)	+0.15 (p=0.0078)
counterfact	edit	ePC s1	300	300	+0.15 (p=0.0081)	+0.15 (p=0.0078)	+1.00 (p=0)	+0.15 (p=0.0078)
counterfact	edit	ePC s2	300	300	+0.15 (p=0.0081)	+0.15 (p=0.0078)	+1.00 (p=0)	+0.15 (p=0.0078)
counterfact	paraphrase	BP s0	600	513	+0.11 (p=0.0059)	+0.47 (p=1e-34)	+0.66 (p=1.2e-76)	+0.17 (p=0.00013)
counterfact	paraphrase	BP s1	600	504	+0.04 (p=0.32)	+0.48 (p=4.1e-36)	+0.60 (p=4.7e-61)	+0.13 (p=0.003)
counterfact	paraphrase	BP s2	600	526	+0.10 (p=0.015)	+0.41 (p=2.8e-25)	+0.72 (p=5.6e-98)	+0.12 (p=0.0056)
counterfact	paraphrase	ePC s0	600	353	+0.03 (p=0.46)	+0.82 (p=2.3e-149)	+0.30 (p=1.2e-13)	+0.22 (p=2.7e-05)
counterfact	paraphrase	ePC s1	600	488	+0.12 (p=0.0042)	+0.53 (p=6.6e-45)	+0.62 (p=8.8e-65)	+0.15 (p=0.0011)
counterfact	paraphrase	ePC s2	600	410	+0.06 (p=0.12)	+0.72 (p=7.7e-96)	+0.43 (p=5.1e-29)	+0.15 (p=0.0019)
counterfact	unseen	BP s0	100	0	—	—	—	—
counterfact	unseen	BP s1	100	1	-0.12 (p=0.23)	+1.00 (p=0)	-0.12 (p=0.23)	—
counterfact	unseen	BP s2	100	0	—	—	—	—
counterfact	unseen	ePC s0	100	0	—	—	—	—
counterfact	unseen	ePC s1	100	1	-0.12 (p=0.23)	+1.00 (p=0)	-0.12 (p=0.23)	—
counterfact	unseen	ePC s2	100	0	—	—	—	—
mquake	edit	BP s0	300	300	-0.29 (p=2.8e-07)	-0.29 (p=3.4e-07)	+1.00 (p=0)	-0.29 (p=3.4e-07)
mquake	edit	BP s1	300	282	-0.27 (p=1.4e-06)	-0.08 (p=0.18)	+0.86 (p=3.2e-91)	-0.30 (p=4e-07)
mquake	edit	BP s2	300	300	-0.29 (p=2.8e-07)	-0.29 (p=3.4e-07)	+1.00 (p=0)	-0.29 (p=3.4e-07)
mquake	edit	ePC s0	300	300	-0.29 (p=2.8e-07)	-0.29 (p=3.4e-07)	+1.00 (p=0)	-0.29 (p=3.4e-07)
mquake	edit	ePC s1	300	300	-0.29 (p=2.8e-07)	-0.29 (p=3.4e-07)	+1.00 (p=0)	-0.29 (p=3.4e-07)
mquake	edit	ePC s2	300	300	-0.29 (p=2.8e-07)	-0.29 (p=3.4e-07)	+1.00 (p=0)	-0.29 (p=3.4e-07)
mquake	paraphrase	BP s0	300	215	-0.12 (p=0.034)	+0.50 (p=2.6e-20)	+0.60 (p=1.1e-30)	-0.33 (p=7.8e-07)
mquake	paraphrase	BP s1	300	258	-0.22 (p=9.6e-05)	+0.13 (p=0.024)	+0.76 (p=1.1e-56)	-0.36 (p=3e-09)
mquake	paraphrase	BP s2	300	248	-0.24 (p=2.9e-05)	+0.22 (p=0.00016)	+0.70 (p=1.1e-45)	-0.37 (p=1.2e-09)
mquake	paraphrase	ePC s0	300	212	-0.15 (p=0.01)	+0.51 (p=1.7e-21)	+0.57 (p=3.9e-27)	-0.35 (p=2.3e-07)
mquake	paraphrase	ePC s1	300	217	-0.12 (p=0.034)	+0.47 (p=6.2e-18)	+0.61 (p=3.6e-32)	-0.37 (p=2e-08)
mquake	paraphrase	ePC s2	300	208	-0.12 (p=0.031)	+0.52 (p=7.4e-22)	+0.57 (p=2.6e-27)	-0.39 (p=5e-09)
mquake	unseen	BP s0	100	0	—	—	—	—
mquake	unseen	BP s1	100	0	—	—	—	—
mquake	unseen	BP s2	100	0	—	—	—	—
mquake	unseen	ePC s0	100	0	—	—	—	—
mquake	unseen	ePC s1	100	0	—	—	—	—
mquake	unseen	ePC s2	100	0	—	—	—	—
zsre	edit	BP s0	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	edit	BP s1	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	edit	BP s2	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	edit	ePC s0	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	edit	ePC s1	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	edit	ePC s2	300	300	-0.22 (p=0.00015)	-0.22 (p=0.00016)	+1.00 (p=0)	-0.22 (p=0.00016)
zsre	paraphrase	BP s0	300	293	-0.16 (p=0.0054)	-0.09 (p=0.11)	+0.97 (p=2.9e-177)	-0.17 (p=0.003)
zsre	paraphrase	BP s1	300	298	-0.18 (p=0.0022)	-0.15 (p=0.0092)	+0.99 (p=3e-239)	-0.17 (p=0.0027)
zsre	paraphrase	BP s2	300	294	-0.16	-0.11	+0.97	-0.17

dataset	model	n (locality-type probes)	A			B			A∩B	P(A)	P(A B)	tail lift	A	B
counterfact	BP s0	1150	0	4	0	0.000	0.000	—	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
counterfact	BP s1	1150	0	6	0	0.000	0.000	—	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
counterfact	BP s2	1150	0	1	0	0.000	0.000	—	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
counterfact	ePC s0	1150	1	11	1	0.001	0.091	104.55	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
counterfact	ePC s1	1150	4	21	4	0.003	0.190	54.76	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
counterfact	ePC s2	1150	0	5	0	0.000	0.000	—	any locality deterioration (fewer than 10 % of probes deteriorate; most abstain)	any nonzero write (fewer than 10 % fire)				
mquake	BP s0	810	2	3	2	0.002	0.667	270.00	any locality de-	any nonzero write (fewer than				

9. MQuAKE reasoning and sequential behaviour

MQuAKE's registered composition endpoint teaches each case's dependency edits from a fresh start state and asks all three multi-hop questions (all-question success; any-question and question-level rates are additional, not replacements). The Stage-4 v5 reader scored 0/80 all-question and 2/240 question-level at 300 edits in realization 0 [**record**]. The same aggregation applied to this study's six readers (family EXT) and to the fifteen Stage-4 cells (context) is below; `frozen_question_*` are the cap-off (frozen) exact rates on the same questions recorded by the endpoint itself. A failed multi-hop question is not by itself a cap failure, since GPT-2 small may lack the compositional step: the frozen model answers none of the questions with either the old or the new answer.

model	family	stratum	cases	questions	question accuracy	any-question success	all-question success	old answer reappeared	frozen: new exact	frozen: old exact
BP reader s0	EXT	all	80	240	0.0 % (0/240)	0.0 % (0/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
BP reader s0	EXT	hops=2	61	183	0.0 % (0/183)	0.0 % (0/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
BP reader s0	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
BP reader s0	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
BP reader s0	EXT	n_edits=1	78	234	0.0 % (0/234)	0.0 % (0/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
BP reader s0	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	hops=2	61	183	1.1 % (2/183)	1.6 % (1/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	n_edits=1	78	234	0.9 % (2/234)	1.3 % (1/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
BP reader s1	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	hops=2	61	183	1.1 % (2/183)	1.6 % (1/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	n_edits=1	78	234	0.9 % (2/234)	1.3 % (1/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
BP reader s2	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	all	80	240	0.0 % (0/240)	0.0 % (0/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	hops=2	61	183	0.0 % (0/183)	0.0 % (0/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	n_edits=1	78	234	0.0 % (0/234)	0.0 % (0/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
ePC reader s0	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	all	80	240	0.4 % (1/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	hops=2	61	183	0.5 % (1/183)	1.6 % (1/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	n_edits=1	78	234	0.4 % (1/234)	1.3 % (1/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
ePC reader s1	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	hops=2	61	183	1.1 % (2/183)	1.6 % (1/61)	0.0 % (0/61)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	hops=3	16	48	0.0 % (0/48)	0.0 % (0/16)	0.0 % (0/16)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	hops=4	3	9	0.0 % (0/9)	0.0 % (0/3)	0.0 % (0/3)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	n_edits=1	78	234	0.9 % (2/234)	1.3 % (1/78)	0.0 % (0/78)	0.0 %	0.0 %	0.0 %
ePC reader s2	EXT	n_edits=2	2	6	0.0 % (0/6)	0.0 % (0/2)	0.0 % (0/2)	0.0 %	0.0 %	0.0 %
v5_stage4:0-100	S4	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
v5_stage4:0-101	S4	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
v5_stage4:0-102	S4	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
v5_stage4:0-103	S4	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
v5_stage4:0-104	S4	all	80	240	0.8 % (2/240)	1.2 % (1/80)	0.0 % (0/80)	0.0 %	0.0 %	0.0 %
v5_stage4:1-100	S4	all	84	252	0.4 % (1/252)	1.2 % (1/84)	0.0 % (0/84)	0.0 %	0.0 %	0.0 %
v5_stage4:1-101	S4	all	84	252	0.4 % (1/252)	1.2 % (1/84)	0.0 % (0/84)	0.0 %	0.0 %	0.0 %
v5_stage4:1-102	S4	all	84	252	0.4 % (1/252)	1.2 % (1/84)	0.0 % (0/84)	0.0 %	0.0 %	0.0 %
v5_stage4:1-103	S4	all	84	252	0.4 % (1/252)	1.2 % (1/84)	0.0 % (0/84)	0.0 %	0.0 %	0.0 %
v5_stage4:1-104	S4	all	84	252	0.4 % (1/252)	1.2 % (1/84)	0.0 % (0/84)	0.0 %	0.0 %	0.0 %
v5_stage4:2-100	S4	all	88	264	0.0 % (0/264)	0.0 % (0/88)	0.0 % (0/88)	0.0 %	0.0 %	0.0 %
v5_stage4:2-101	S4	all	88	264	0.0 % (0/264)	0.0 % (0/88)	0.0 % (0/88)	0.0 %	0.0 %	0.0 %
v5_stage4:2-102	S4	all	88	264	0.0 % (0/264)	0.0 % (0/88)	0.0 % (0/88)	0.0 %	0.0 %	0.0 %
v5_stage4:2-103	S4	all	88	264	0.0 % (0/264)	0.0 % (0/88)	0.0 % (0/88)	0.0 %	0.0 %	0.0 %
v5_stage4:2-104	S4	all	88	264	0.0 % (0/264)	0.0 % (0/88)	0.0 % (0/88)	0.0 %	0.0 %	0.0 %

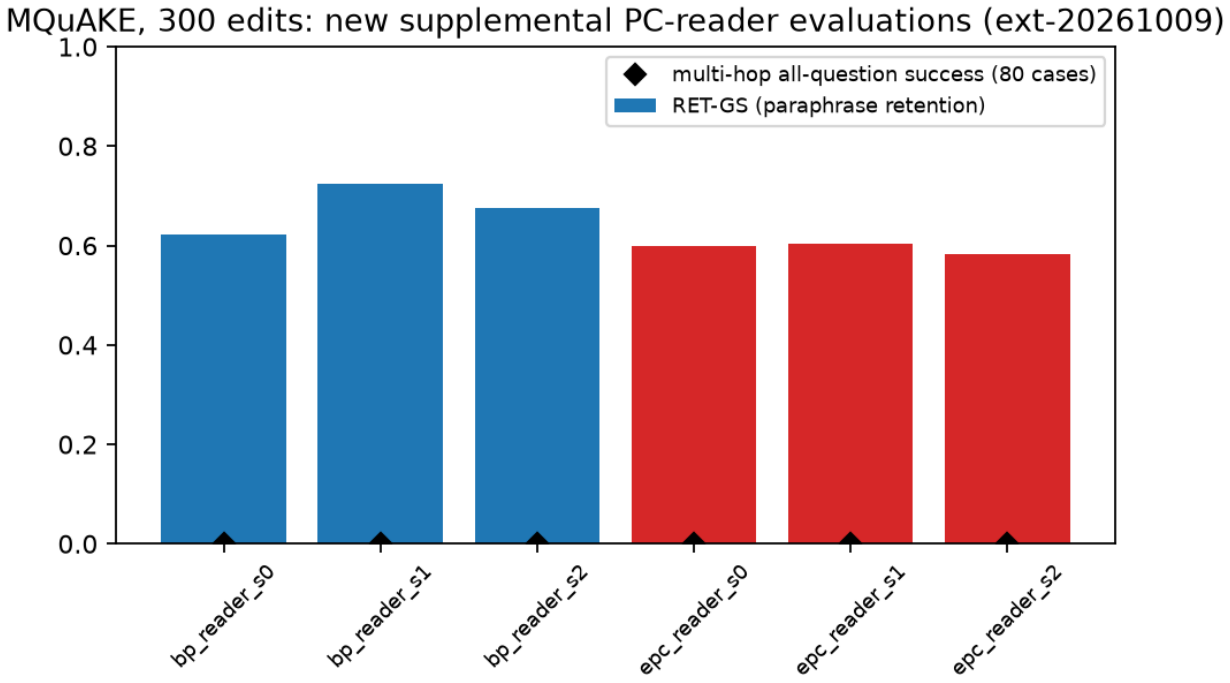


Figure 10: MQuAKE PC-reader evaluations

Sequential behaviour along the real teaching order (order 100) [**recomputed**] from the per-edit records:

dataset	reader	edits	immediate ES	immediate paraphrase score	ES failures	consecutive failures	ACF(gs) lag 1	lag 2	lag 3
counterfact	bp s0	300	100.0 %	83.8 %	0	0	0.08	-0.06	-0.07
counterfact	bp s1	300	99.0 %	82.7 %	3	0	-0.02	-0.01	-0.09
counterfact	bp s2	300	100.0 %	84.8 %	0	0	-0.01	-0.08	0.01
counterfact	epc s0	300	100.0 %	56.3 %	0	0	-0.04	0.08	-0.05
counterfact	epc s1	300	100.0 %	78.3 %	0	0	0.01	-0.03	-0.17
counterfact	epc s2	300	100.0 %	66.5 %	0	0	-0.09	0.08	0.01
mquake	bp s0	300	100.0 %	62.0 %	0	0	-0.02	-0.00	-0.01
mquake	bp s1	300	91.0 %	72.3 %	27	3	0.02	-0.00	0.05
mquake	bp s2	300	100.0 %	67.0 %	0	0	0.02	-0.06	-0.02
mquake	epc s0	300	100.0 %	58.7 %	0	0	0.02	-0.04	-0.03
mquake	epc s1	300	100.0 %	59.0 %	0	0	0.02	-0.02	-0.02
mquake	epc s2	300	100.0 %	57.3 %	0	0	0.02	-0.03	-0.07
zsre	bp s0	300	100.0 %	97.3 %	0	0	-0.03	-0.03	-0.03
zsre	bp s1	300	100.0 %	99.3 %	0	0	-0.01	-0.01	-0.01
zsre	bp s2	300	100.0 %	97.7 %	0	0	-0.02	-0.02	-0.02
zsre	epc s0	300	100.0 %	94.7 %	0	0	-0.06	0.08	-0.06
zsre	epc s1	300	100.0 %	96.7 %	0	0	0.07	0.07	-0.03
zsre	epc s2	300	100.0 %	97.3 %	0	0	-0.03	0.10	-0.03

The ordinary-text assay is not a time series (context resets every window), so no sequential extreme analysis is manufactured there.

10. Connection with nonlinear statistical coupling (Nelson) [new, supplementary]

For every eligible GPD fit in this study the informational-scale condition $-\sigma \cdot \mathrm{d}/\mathrm{dz}\ \ln f(z)$ at $z = \sigma$ equals 1 (for the density $f(z) = \sigma^{1}(1 + \kappa z/\sigma)\hat{}(-1/\kappa - 1)$ this holds identically; checked numerically for 10 fits, all hold: True). The fitted σ is the scale of exceedances above the chosen threshold, not a threshold-free scale of the full loss distribution; shapes and scales are tabulated separately in sections 6.1–6.3. The GPD shape κ is Nelson's nonlinear coupling parameter in the one-dimensional $\alpha = 1$ family; a fitted $\kappa > 0$ gives a modelled survival exponent $1/\kappa$ and must not be confused with the stretching parameter α.

Exploratory coupled-entropy profile (discrete, $\alpha = 1$, k = 1; $q(\kappa) = (1+2\kappa)/(1+\kappa)$; $H_\kappa(p) = [1/\sum p_j\hat{}q - 1]/\kappa$; Shannon at $\kappa = 0$; self-tests: {"point_mass_zero": true, "shannon_at_zero": true, "uniform_closed_form": true, "convergence_to_shannon": true, "rejects_unnormalized": true}), computed on the relation/new-target/subject frequency vectors of each population (`tables/coupled_entropy_profiles.csv`, 60 rows). Increasing κ lowers H_κ for every non-uniform distribution and compresses differences between distributions with many rare categories; e.g. for the CounterFact eligible relation distribution $H_0 = 3.409$

and H_1 = 4.402. This is a sensitivity study of empirical categorical distributions, not a calibrated entropy of the system and not evidence for the paper's universal-entropy theorem; it is kept apart from the HT-17 fits. The earlier bounded κ-surprisal pilot [**record**] is not a test of the coupled-entropy objective.

11. Discussion and conclusions

What the results establish. On GPT-2 small, both reader caps install the taught facts (own-prompt per-token loss ≈ 0.01 nats against ≈ 6 for the frozen model) and are exactly the frozen model where they abstain. The ePC- and BP-trained readers differ in gating, not in the installed corrections (the memory payloads are produced by the same adjoint rule, so own-prompt losses coincide across all six readers). The frozen record's negative CounterFact result for ePC reader training is reproduced in the per-probe losses and persists across difficulty deciles, on the frozen model's hardest cases and in the regression tails; on zsRE the two rules are close. No analysis in this study isolates a rare-event regime in which ePC reader training is reliably better than BP reader training; the measured cost difference ($\approx 100\times$) is unchanged.

Per-dataset paired contrasts at 300 edits (seed-level intervals from 2,000 item-group bootstrap draws; three seeds share one subject population):

- **zsre:** paraphrase-prompt contrast ePC − BP (positive favours ePC), seed mean -0.076 [-0.136, -0.027] nats/token (seed 0: -0.159 [-0.282, -0.055]; seed 1: -0.092 [-0.176, -0.023]; seed 2: 0.021 [-0.048, 0.099]). On the frozen model's hardest 5 % of paraphrases, mean per-token loss falls from 11.37 (frozen) to 0.05 (BP, seed mean) and 0.05 (ePC); exact-match success there is 100.0 % (BP) vs 100.0 % (ePC). ePC-minus-BP deteriorations above 1 nat/token occur on 2.7 %, 2.0 %, 0.7 % of paraphrases (seeds 0–2); ePC is better than BP by more than 1 nat/token on 0.3 %, 0.0 %, 1.0 %.
- **counterfact:** paraphrase-prompt contrast ePC − BP (positive favours ePC), seed mean -1.117 [-1.313, -0.916] nats/token (seed 0: -1.821 [-2.168, -1.447]; seed 1: -0.149 [-0.357, 0.057]; seed 2: -1.379 [-1.666, -1.085]). On the frozen model's hardest 5 % of paraphrases, mean per-token loss falls from 10.42 (frozen) to 2.05 (BP, seed mean) and 2.78 (ePC); exact-match success there is 78.9 % (BP) vs 74.4 % (ePC). ePC-minus-BP deteriorations above 1 nat/token occur on 28.5 %, 8.2 %, 22.3 % of paraphrases (seeds 0–2); ePC is better than BP by more than 1 nat/token on 3.5 %, 4.8 %, 3.3 %.
- **mquake:** paraphrase-prompt contrast ePC − BP (positive favours ePC), seed mean -0.495 [-0.651, -0.347] nats/token (seed 0: -0.090 [-0.299, 0.106]; seed 1: -0.718 [-0.943, -0.502]; seed 2: -0.677 [-0.882, -0.486]). On the frozen model's hardest 5 % of paraphrases, mean per-token loss falls from 10.17 (frozen) to 3.13 (BP, seed mean) and 3.13 (ePC); exact-match success there is 33.3 % (BP) vs 33.3 % (ePC). ePC-minus-BP deteriorations above 1 nat/token occur on 5.3 %, 13.7 %, 13.3 % of paraphrases (seeds 0–2); ePC is better than BP by more than 1 nat/token on 4.3 %, 0.3 %, 0.0 %.

Extremes. The caps' collateral effects on ordinary text are rare and sometimes severe (HT-17 reproduced exactly; new MQuAKE rows in 6.2); the per-probe loss tails are dominated by the frozen model's own difficulty, with GPD shapes near zero. Severe per-probe regressions are concentrated on prompts where a wrong or stale record fired, which is why they correlate with write magnitude rather than with frozen difficulty.

What remains uncertain. One realization, one order, three training seeds; MQuAKE PC-reader results are new and unreplicated; tail shapes are finite-range; the frozen model's exact-match rates are uninformative on these prompts; transfer beyond GPT-2 small is not established. Figures with fewer than 100 exceedances are marked exploratory or insufficient in the tables.

12. Reproducibility appendix

- **Code:** `pc_cap/aw/extremes/` at revision `a3727cf5e248`; tests `aw/tests/test_extremes_stats.py` (CPU) and `aw/extremes/validate_gpu.py` (GPU; record in `validation/gpu_checks.json`).
- **Commands:** `results/extremes_analysis/ext-20261009/scripts/run_gpu_chain.sh` (frozen scoring → reader scoring → MQuAKE evaluations → MQuAKE scoring) and `scripts/run_cpu_pipeline.sh` (assemble → paired → tails → datasets → nelson → figures → report). Environment: Python 3.12.15, jax/jaxlib 0.11.1, numpy 2.5.3, scipy 1.18.1, pandas 3.0.5, pyarrow 25.0.1, matplotlib 3.11.1; RTX 5070 (12 GiB), driver 615.71.09.

- **Inputs (hash-checked at load):** sealed recipes `docs/tasks/R1-final-cell-recipes/{61348508…, ce0d0ffc…, 43925e53…}.json` and their payloads; reader artifacts `assets/runs/additional_work/PC-reader/train-*/theta_avg150-30` memory snapshots `…/eval-*/stream/checkpoint-{100,300}.snapshot`; harm vectors `results/additional_work/PC-reader` HT-17 `logs/additional_work/HT-17/snapshot-20261004-complete/report.json`.
- **Outputs:** parquet under `assets/extremes_analysis/ext-20261009/data/` (example manifest = `scores.parquet` metadata columns + `case_populations.parquet`; `scores.parquet`, `generations.parquet`, `paired_cases.parquet`, `dataset_statistics.parquet`, `tail_fits.parquet`, `bootstrap_results.parquet`, `paired_gains.parquet`, `difficulty_decil` CSV tables under `tables/`; figures under `assets/extremes_analysis/ext-20261009/figures/` with `captions.json`; raw per-probe records as JSONL chunks with a hash manifest under `assets/extremes_analysis/ext-20261009/checkpoin` MQuAKE evaluation receipts under `results/extremes_analysis/ext-20261009/mquake_eval/`.
- **Data dictionary:** `reports/data_dictionary.md`. **Artifact registry:** `manifest.json` (72 artifacts with SHA-256).
- **Formulas:** §3 (losses, margins), §6 (harm, ES99+, GPD), §7 (gains, D, CVaR, deciles), §10 (coupled entropy). Natural logarithms throughout.